

Microscopic Dynamics of the Space-Time Membrane: A Conservative Elastic Framework for Emergent Structure

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Abstract

We develop a conservative microscopic formulation of the Refined Space-Time Membrane (RSTM) model in $3 + 1$ dimensions in which six result groups—**Structure, Scale, Particle, Family, Quantum and Geometry**—arise through microscopic dynamics, selection and conservative reduction. A real constitutive coordinate relaxes intermediate-gradient stiffness while preserving a positive microscopic parent, and a two-real-dimensional internal coordinate with primitive C_3 symmetry generates three symmetry-related orientations. Eliminating stable constitutive fluctuations yields a negative Schur-complement correction to fourth-order stiffness with correlated positive sixth-order regulation, producing a stable finite-wave-number corridor. Persistent localisation instead arises in a conservative two-face sector with fixed Noether charge, where reciprocal constitutive coupling supports a stable three-dimensional one-orbital branch below the dispersed same-charge continuum. The same architecture admits a stable flavour-sensitive fixed-charge branch without inserting a family mass matrix or microscopic complex Yukawa phase. Positive-frequency reduction yields Schrödinger-like evolution, while conservative analyser dynamics gives squared-overlap channel weights. At collective spatial level, incompatible strain reproduces the linearised spatial Einstein tensor and relabelling symmetry fixes the quadratic curvature coefficients in the Einstein ratio. Physical coefficient descent, local gauge dynamics, universal Born preparation statistics and four-dimensional gravitational closure remain open tests.

Keywords: Space-Time Membrane; Conservative Field Theory; Higher-Gradient Elasticity; Fixed-Charge Localisation; Family/Flavour Structure; Emergent Quantum Dynamics; Microscopic Constitutive Dynamics; Emergent Gravity; Lorentz Symmetry Emergence; Ultraviolet Regularisation

1. Introduction

Modern physics rests on two exceptionally successful but conceptually different descriptions of nature. General Relativity (GR) describes gravitation through dynamical spacetime geometry, whereas quantum mechanics and quantum field theory describe microscopic phenomena through complex amplitudes, operator algebras and probabilistic measurement rules. Both frameworks have been tested to remarkable precision, yet their combination remains incomplete, particularly where quantum structure, strong gravitation and ultraviolet physics meet. The continuing difficulty is not only to construct a mathematically consistent high-energy theory, but also to explain why the familiar low-energy structures of quantum theory and relativity take the forms that they do.

The Space-Time Membrane (STM) programme explores a different starting point. Spacetime is treated as a four-dimensional elastic substrate with a complementary conjugate-domain structure, and familiar physical behaviour is sought as collective dynamics of that substrate rather than inserted as a set of independent fundamental laws. The earlier refined STM/RSTM formulation

[1] developed this idea through an explicit higher-gradient scalar equation containing second-, fourth- and sixth-order spatial operators. That formulation was deliberately broad: it tested whether one elastic framework could support gravitational, quantum-like, gauge-like, matter and strong-field structures. In doing so, it also exposed an important limitation. Several quantities that appeared as coefficients or effective ingredients of the scalar equation had not themselves been derived from a deeper microscopic action.

The present paper moves beneath that effective description. The higher-gradient scalar equation is retained, but it is no longer treated as the complete microscopic law. Instead, it appears downstream of a larger conservative system in which local constitutive degrees of freedom respond to the membrane state and are subsequently relaxed, projected or eliminated. The central organising principle is therefore

$$S_{micro} \rightarrow \text{coupled Euler - Lagrange dynamics} \rightarrow \text{dynamically selected structure} \rightarrow$$

$$\Gamma_{eff} \rightarrow \text{effective law} \rightarrow \text{observable}.$$

We refer to this recurring sequence as emergence by conservative reduction. The microscopic system remains conservative at the declared level; apparently dissipative, non-local or probabilistic structures may arise only after information carried by unresolved constitutive, environmental or reference degrees of freedom has been removed from the description.

Why additional microscopic coordinates are required

The introduction of additional constitutive variables is not motivated by adding freedom to the earlier RSTM equation. It follows from two limitations exposed by the post-effective reconstruction. First, the scalar coordinate u was being asked simultaneously to describe membrane deformation and to account for changes in its own effective constitutive response. In particular, finite-scale structure requires sufficient intermediate-gradient softening while the unreduced microscopic theory must remain positive and stable. This poses the microscopic question

Can a healthy positive system generate $\Delta K_4 < 0$ without postulating negative microscopic stiffness?

For a retained scalar fluctuation coupled to a healthy hidden coordinate with positive Hessian H , constitutive relaxation gives the Schur-complement response

$$A_{eff} = A - CH^{-1}C^T.$$

Because $CH^{-1}C^T$ is positive semidefinite, relaxation necessarily softens the retained response [2]. The minimal additional variable required for the leading scalar curvature coupling is therefore one real common constitutive coordinate,

$$s \in \mathbb{R},$$

entering, for example, through

$$\frac{1}{2}K_4(s)(\Delta u)^2.$$

On a curvature-populated branch this produces the correlated response derived below,

$$\Delta K_4 < 0, \quad \Delta K_6 = -\downarrow_s^2 \Delta K_4 > 0.$$

The second requirement is independent. A single common coordinate does not provide a natural internal orientation structure capable of generating three equivalent states without inserting a three-well or three-field construction by hand. The smallest real internal space supporting a non-trivial cyclic rotation is two-dimensional. Writing this relative sector as

$$w = w_1 + iw_2 \in \mathbb{C}$$

is therefore the minimal choice. The theory adopts the exact constitutive symmetry

$$w \rightarrow e^{2\pi i/3} w$$

as a primitive postulate. Its lowest anisotropic invariant, $Re(w^3)$, then generates three symmetry-related orientations. The number three is consequently a derived property of the admitted C_3 constitutive structure rather than the consequence of introducing three primitive generation fields.

The resulting minimal constitutive decomposition is

$$\boxed{1_{common} \oplus 2_{relative}}$$

with s the leading $M1$ -bright response and the two real components of w symmetry-protected from a generic linear scalar response at $w = 0$. The two sectors therefore solve different microscopic problems and are kept distinct from the later spin, chirality, charge and gravitational constructions.

Microscopic RSTM. The preceding requirements lead to the minimal core variables: a scalar membrane coordinate u , a real common constitutive coordinate s , and a complex internal redistribution coordinate w ,

$$\chi_{micro} = (u, s, w), \quad s \in \mathbb{R}, \quad w \in \mathbb{C}.$$

The coordinate s is the $M1$ -bright common response selected by the scalar constitutive problem, whereas w carries the two-real-dimensional relative sector required for internal orientation. At the symmetric point $w = 0$, exact C_3 symmetry makes this relative doublet linearly dark to the universal scalar constitutive response, giving the natural decomposition

$$(s, Rew, Imw) \simeq 1_{M1} \oplus 2_{relative}.$$

$$w \rightarrow e^{2\pi i/3} w.$$

This C_3 symmetry is adopted as a primitive constitutive postulate. It is not inferred from the observed number of particle generations. The mathematical fact that a C_3 -invariant anisotropy can possess three symmetry-related minima is not itself claimed as novel. The RSTM proposal is instead that the smallest relative constitutive sector required independently by the microscopic ontology carries this symmetry, so that a three-state family-like multiplicity arises without introducing three independent primitive family fields. The physical mass hierarchy, flavour mixing and CP structure remain separate downstream problems.

The present work goes further than the multiplicity result while preserving this distinction. Exact C_3 symmetry alone leaves the three orientations equivalent and therefore cannot supply a hierarchy or non-trivial mixing. Family differentiation is instead sought relationally after the three states have formed. Domain-resolved coordinates w_p and w_m provide an independently responding conjugate reference, with $Y = Im(w_p w_m)$ the minimal real fold-odd relational bilinear. Coupling this response to the existing τ_2 face/ u coherence through the positive-square matter parent, and transmitting its radial response through the common s sector, provides an action-class route to family-conditioned particle profiles, inter-family transition kernels and

CP-capable effective mass operators. No observed fermion mass, CKM or PMNS element, or fitted microscopic complex phase is used to select the branch.

Prior-art and novelty boundary. Discrete cyclic family symmetries, circulant flavour textures, three-state flavour constructions and Z_3 -based generation-triplication mechanisms have substantial precedent [3–7]. Exact cyclic or circulant limits may possess a common diagonalising basis and therefore require additional sector misalignment or symmetry breaking to generate general physical mixing [3–5]. RSTM does not claim those mathematical facts as novel. The narrower proposal tested here is their different microscopic provenance and dynamical ordering: three equivalent states first arise in the independently motivated constitutive w sector; an independently responding conjugate-domain coordinate then supplies the relational fold-odd discriminator; that response enters the already required positive-square face dynamics; and the resulting flavour-sensitive fixed-charge branch is selected and stability-tested before any phenomenological family mass matrix is constructed. In the literature examined for the present comparison, we did not identify a construction combining this complete sequence. Physical masses, CKM/PMNS matrices and CP violation remain downstream blind tests. The additional variables do not represent extra spacetime dimensions. The theory remains in 3+1 dimensions,

$$x^\mu = (t, x^1, x^2, x^3) \in \mathbb{R}^{1,3}.$$

with s , w and the later face/frame variables defined internally over the same spacetime. This distinction is important: the constitutive family doublet carried by w , the reduced two-component spin fibre, the two-face occupation space and the collective gravitational tensor variables are different objects with different dynamical roles.

Methodological reversal. The current formulation changes the direction in which parameters are introduced. The observed values of Newton's constant, the quantum of action and the Planck length are not used as microscopic inputs from which the same quantities are subsequently recovered. Instead, the microscopic action is specified first; its Euler-Lagrange equations, stable branches and reduced kernels determine effective coefficients; and only then are the resulting dimensional scales compared with observation. Numerical values used in the calculations below therefore fall into two distinct categories. Benchmark parameters test existence, stability and mechanism, whereas physical parameters must ultimately descend from the microscopic action. A successful benchmark is evidence that a mechanism is available; it is not, by itself, a prediction of nature's coefficient values.

Structure and finite-scale selection. The first major result follows from allowing the common constitutive coordinate to relax on a curvature-populated branch. A healthy positive-energy internal degree of freedom produces a negative Schur-complement correction to the relaxed fourth-order scalar response while its finite response length simultaneously generates a positive sixth-order correction,

$$\Delta K_4 < 0, \quad \Delta K_6 > 0.$$

Intermediate-scale softening and short-scale regulation therefore arise from the same microscopic response rather than from independently chosen signs. The resulting effective kernel can remain positive for all physical modes while developing a preferred non-zero wave number over a finite dimensionless corridor. Thus the mechanism selects a characteristic scale through competition between tension, relaxed intermediate-gradient stiffness and sixth-order regulation without requiring microscopic instability. Finite-wavenumber minima and modulated phases have established precedents in Brazovskii- and Swift-Hohenberg-type systems [8,9]; the distinction here is that the effective RSTM coefficients arise from conservative constitutive relaxation rather than being prescribed phenomenologically.

This result also changes the interpretation of the earlier RSTM scalar equation. The characteristic 2-4-6 derivative hierarchy survives, but its effective coefficients are correlated outputs of constitu-

tive dynamics. The finite-wave-number minimum identifies a preferred structured-response scale; it is not itself used as proof of a persistent particle. Direct analysis of the positive scalar operator shows that the scale-selection problem and the matter-binding problem must be kept distinct.

Matter in RSTM. Persistent localisation is instead developed through an explicit conservative two-face occupation sector. The face action is constructed from a globally positive square-form mass parent and is postulated to possess an exact common face-phase symmetry. Noether's theorem then supplies a conserved charge Q_F [10]. On the occupied lower branch, reciprocal coupling to the common constitutive coordinate creates a self-consistent mass well, and the physical localisation problem is posed at fixed Q_F rather than as an unconstrained scalar defect. Fixed-charge non-topological localisation has a well-established precedent in Q-ball theory [11]. Conserved Noether charge can also coexist with non-trivial topology, as in Q-lump constructions [12], while spinning Q-balls provide explicit charge-supported rotating solitons in $3+1$ -dimensional Minkowski space [13]. Higher-spatial-derivative Lifshitz theories likewise admit topological Q-ball solutions [14]. The present work therefore does not claim novelty for fixed-charge localisation, the coexistence of Noether charge and topology, rotating charge-supported solitons, or higher-gradient topological soliton structure as such. The narrower RSTM distinction is the dynamically derived division of roles: constitutive relaxation selects a preferred finite-wave-number structural scale but is shown not by itself to bind the particle, whereas a separate positive-square two-face sector supplies reciprocal fixed- Q_F localisation.

For a finite representative coefficient and charge interval, the coupled equations admit a localised relative equilibrium containing one preferred spatial orbital. The branch lies below the dispersed continuum carrying the same charge and satisfies the fixed-charge variational relation

$$\frac{dE}{dQ_F} = \omega.$$

The circular solution remains stationary under direct conservative evolution, survives finite radial perturbation, and exhibits no growing lower/common mode through the tested non-radial angular sectors. The expected phase and translation symmetry modes are recovered, while the dormant complementary-face sector possesses an explicit open Schur-coercive stability corridor. These calculations establish an existence-class particle result: a conservative action of the stated type can support a three-dimensional, one-orbital, fixed-Noether-charge localised carrier. The physical face coefficients and the absolute elementary charge/action carried by nature remain to be derived independently.

Spin, chirality and exchange. The local occupation amplitude is not required to carry the complete topological texture. Earlier same-field constructions tended either to push winding into the negligible-density exterior or to damage the localised branch when the winding was forced into the core. The current formulation therefore introduces a distinct normalised phase frame and attaches it covariantly to the occupation field. Pure frame orientation is then isospectral to the underlying occupation operator: binding remains an amplitude/mass effect, whereas the phase frame carries the rotational topology.

Topological mechanisms for half-integer rotational behaviour and fermionic soliton quantisation have established precedent in Finkelstein-Rubinstein-type constructions [15], while charged spinning solitons are independently known from Q-ball theory [13]. The novelty claimed here is therefore not that topology can support spinorial behaviour, nor that a charge-supported soliton can rotate. It is the specific separation realised by the RSTM construction: the localised occupation amplitude is first bound independently at fixed Q_F , after which a distinct normalised degree-one frame is attached covariantly so that its orientation carries the spin-cover topology without changing the binding spectrum.

The resulting reduced one-particle sector contains a rank-two spin fibre with the characteristic double-cover response under a 2π rotation. This rotational sign is kept logically separate from particle exchange. Topological and configuration-space relations between rotation and exchange have established precedents [15,16]; the present construction deliberately derives the one-particle

rotation sign and two-particle exchange character separately. An independent two-particle exchange construction gives an odd relative exchange character at the reduced/topological level; conditional on physical selection of that odd one-particle sector, antisymmetric exterior-Fock construction yields the usual mode canonical anticommutation relations. Compulsory microscopic selection of the odd sector, its promotion to a local Spin(1,3)-covariant field, particle/antiparticle completion and graded microcausality remain open.

Chirality is treated separately again. The two-face/orientation structure supplies a projector architecture compatible with left-handed visible weak currents and a chirality-reversed complementary sector, but face reflection is not identified with fermion parity or with the two-particle exchange sign. Likewise, local internal-frame covariance supplies a structural gauge-connection architecture, while the physical local Yang-Mills dynamics, kinetic normalisation and couplings have not yet been derived from the microscopic action. This separation prevents several superficially similar two-component or two-face structures from being used interchangeably.

Quantum-like dynamics. Complex evolution is not introduced at the microscopic level as an independent stochastic postulate. A stable real conservative mode possesses positive- and negative-frequency branches; selecting the positive-frequency sector equips the real canonical phase space with its natural complex structure. For the circular face carrier, the common phase symmetry also provides an exact action variable, and the phase, spectral and symplectic definitions of the classical carrier action coincide on that branch. The corresponding reduced evolution has the Schrödinger-like form

$$iJ_* \partial_t \psi = H_{eff} \psi.$$

where J_* denotes the elementary action scale required by the quantum interface. The structural action variable is therefore identified, but its absolute universal normalisation is not. In particular, $J_* = \hbar$ is not imposed microscopically. A global rescaling of the classical action leaves the Euler-Lagrange trajectories unchanged while rescaling momenta, energies and actions, so the dimensional quantum of action must be fixed by an additional microscopic occupancy or interface condition and tested against $\hbar$ only afterwards.

Measurement is treated at the same reduced level. Linear projection onto detector channels followed by quadratic energy response produces Born-form weights proportional to squared overlaps [17]. A deterministic threshold detector can then amplify one channel into a persistent macroscopic record, while environmental or unresolved detector variables generate memory, decoherence and effective damping after reduction. The closed microscopic theory itself does not require a universal Rayleigh term. The remaining statistical problem is sharper than in the earlier formulation: a universal Born rule requires a derivation of the physical preparation measure over unresolved microscopic detector and state variables for arbitrary measurement contexts. The present calculation establishes the quadratic channel response and representative deterministic recording mechanism, not yet that universal preparation theorem.

Gravity in microscopic RSTM. The gravitational interpretation is also reorganised. Ordinary compatible strain of a displacement field has vanishing Saint-Venant incompatibility and is therefore treated as a change of spatial reference or ruler rather than as physical curvature. Curvature is carried instead by an independent incompatible collective strain constructed from scalar, vector and traceless-quadrupole moments of the microscopic state. The scalar and quadrupole sectors together supply the six independent components required for a collective spatial metric, and at linear order the incompatibility tensor is the spatial Einstein tensor [18]. Spatial relabelling symmetry then imposes a Ward identity on the effective quadratic curvature action and fixes the relative coefficients to the Einstein ratio rather than leaving four independent two-derivative parameters to be tuned. This is a structural spatial result, not yet a complete microscopic derivation of four-dimensional GR. The physical common transverse-traceless pole, its residue and propagation cone, the metric-normalised gravitational coupling, temporal Fierz-Pauli recombination [19] and the nonlinear hypersurface-deformation algebra [20] remain downstream

calculations. The construction deliberately allows the microscopic substrate to retain a preferred clock or rest-frame memory in higher-derivative ultraviolet terms, provided the leading infrared spatial reference dependence is erased and the required Lorentz/Einstein structure emerges in the physical long-wavelength sector.

UV control and strong-field structure. The conservative microscopic action remains second order in time while containing spatial derivatives through sixth order. Its high-momentum behaviour therefore has Lifshitz-type $z = 3$ scaling [21]. For the selected finite scalar/constitutive truncation, weighted power counting [22] is super-renormalisable at the declared scope, and the stable free scalar Euclidean covariance is reflection positive with respect to the preferred Euclidean time. Because the microscopic action remains second order in time, the higher spatial derivatives do not introduce ordinary higher-time-derivative Ostrogradsky modes [23]. These statements do not establish complete coupled RSTM renormalisability or an all-scale Lorentz-invariant Osterwalder-Schrader reconstruction [24,25]. The fundamental fermion and gauge ultraviolet operators, collective gravity completion, radiative protection of the infrared Lorentz surface and ultimate microscopic causal cut-off remain open.

The same sixth-order positive energy provides a strong energetic resistance to zero-width localisation. Together with the binding and constitutive terms, it supports finite-width regulated cores in the selected truncations. A finite microscopic core should not, however, be equated automatically with a regular black-hole spacetime: transfer of microscopic regularity to the collective metric, bounded curvature invariants, horizon structure and geodesic completeness require separate derivations. The retained core calculation also exhibits a leading area-law scaling at its stated effective level. This is distinct both from the conditional Bekenstein-Hawking/Wald entropy relation [26–28] that would follow from a correctly normalised infrared Einstein-Hilbert action and from the still-open microscopic count of physical horizon states.

Cross-sector normalisation. The new parameter programme keeps the microscopic length, action and gravitational sectors independent until their own dynamical calculations have been completed. A normalisation-independent dimensionless combination can then be formed from the microscopic crossover length ℓ_{**} , the elementary action scale J_{**} and the emergent gravitational coupling G_{RSTM} . Its purpose is not to insert the Planck relation by definition, but to provide a blind cross-sector test of whether independently derived quantities converge on it. The conventional Planck scale is therefore a possible closure result of the theory rather than its assumed microscopic ruler.

Relation to contemporary substrate-first approaches. Recent work has independently explored the possibility that relativistic, gauge and matter-like structures may emerge from deterministic elastic substrates. Molzberger develops a deterministic four-dimensional brane-lattice construction in which spacetime, gauge structure and localised matter are sought as emergent properties of the underlying lattice, while Jiang develops Lorentz and gauge symmetries from dynamical constraints on excitations of a continuous elastic substrate [29,30]. Singh's Quantum Elastic Spacetime Theory (QuEST) likewise treats spacetime as an elastic medium with strain-based gravitational dynamics and high-density singularity regulation [31], but employs different strain/configuration variables and a stochastic rearrangement mechanism rather than the conservative constitutive reduction developed here. A particularly close contemporary gravitational comparison is provided by Kulkarni's intrinsic D_4 -lattice construction [32]. There, compatible vertex-displacement strains are identified with linearised diffeomorphism directions and carry no curvature, whereas incompatible intrinsic edge-length modes carry the gravitational degrees of freedom. The resulting Regge kinetic operator is shown, on the stated D_4 subdivision, to coincide with the linearised Einstein/Fierz-Pauli operator. This establishes that the broad association between compatible deformation and gauge redundancy, incompatible deformation and curvature, and the resulting linearised Einstein structure is not unique to RSTM. The microscopic constructions are nevertheless materially different. Kulkarni obtains the operator from an intrinsic discrete Regge lattice and its edge-length Hessian; RSTM instead constructs

a continuous collective strain from scalar, vector and traceless-quadrupole moments of its microscopic state and constrains the general collective two-derivative kernel through spatial-reference redundancy. These approaches share the broad substrate-first objective of deriving familiar effective structure from a smaller classical ontology, but their microscopic mechanisms differ materially from those developed here. In RSTM, the characteristic scalar hierarchy arises through constitutive relaxation, persistent matter is supplied separately by the fixed-Noether-charge two-face sector, and spatial gravitational structure is carried by incompatible collective strain. The relevant proximity is therefore at the level of research programme rather than identity of microscopic ontology or derivation.

The microscopic reconstruction produces six principal result groups that organise the remainder of the paper:

Structure		Scale		Particle		Family		Quantum		Geometry
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Structure: the familiar RSTM 2–4–6 scalar hierarchy is recovered downstream of positive microscopic constitutive dynamics, with correlated fourth- and sixth-order responses rather than independently prescribed effective coefficients.

Scale: those correlated coefficients admit a stable finite-wave-number corridor,

$$-2 < \beta_c < -\sqrt{3},$$

which selects a preferred non-zero structural scale without microscopic instability.

Particle: scale selection is shown not to be sufficient for binding. Persistent localisation instead arises through an independent conservative fixed-Noether-charge face mechanism, with a representative one-orbital branch lying below the dispersed same-charge continuum and surviving the stated stability tests.

Family: the three C_3 -related internal orientations acquire a relational face/domain response through the single-fold Y -channel. A coupled flavour-sensitive fixed-charge branch exists and remains stable in the tested nonlinear and co-rotating controls; physical family masses and mixing remain blind downstream predictions.

Quantum: positive-frequency reduction of the real conservative dynamics supplies the complex Schrödinger-like structure and identifies a classical carrier action, while leaving its universal physical normalisation $J_* = \hbar$ as a blind downstream test.

Geometry: incompatible collective strain reproduces the linearised spatial Einstein tensor, while spatial relabelling symmetry fixes the leading quadratic curvature coefficients in the Einstein ratio

$$1 : -2 : 2 : -2.$$

These results occur at different claim levels — derived, existence-class, reduced or structural as specified below — but together establish the central result of the paper: several apparently different effective structures arise through the same sequence of microscopic dynamics, physical selection and conservative reduction.

The resulting action-first hierarchy and the different claim levels of its six principal result groups are summarised in Figure 1.

Figure 1. Schematic hierarchy of the RSTM construction and claim levels

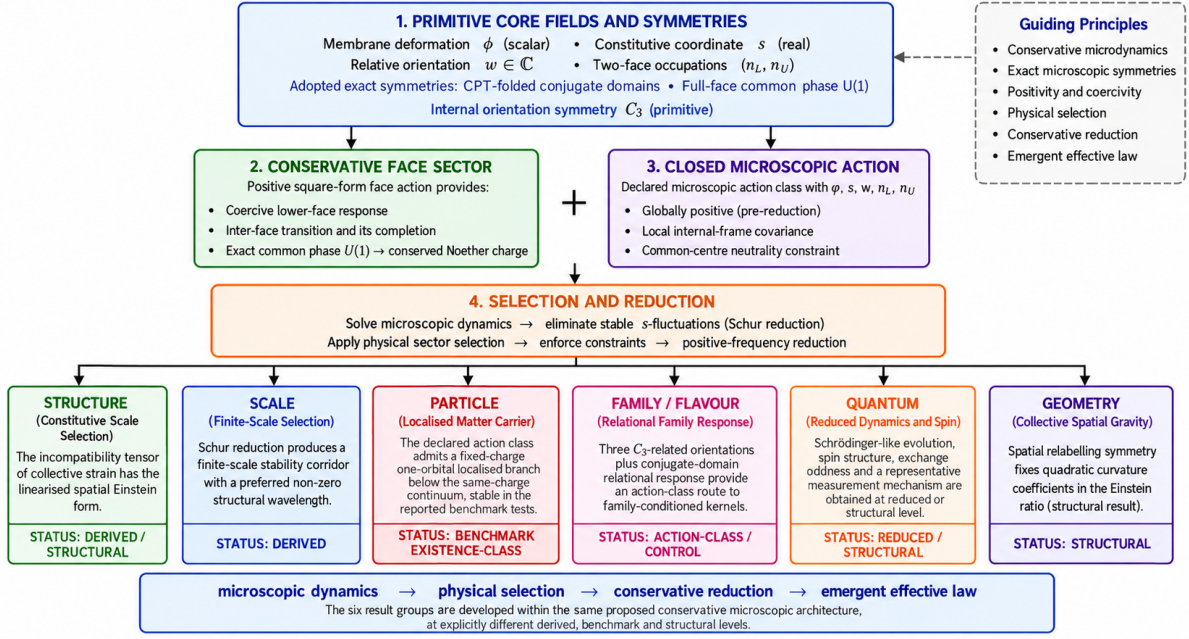


Figure 1: Unified architecture: from primitive core action to emergent structures. The minimal microscopic core comprises the scalar substrate coordinate u , the real common constitutive coordinate s and the complex internal redistribution coordinate w . Exact C_3 symmetry acts on w , while the conservative two-face occupation sector supplies the positive-square matter parent and an exact common phase symmetry. The closed action is varied before reduction; stable or otherwise admissible branches are then selected and unresolved constitutive, reference or environmental detail may be eliminated or projected. The resulting descent organises the six principal result groups—Structure, Scale, Particle, Family/Flavour, Quantum and Geometry—whose status labels distinguish derived/structural, derived, benchmark existence-class, action-class/control, reduced/structural and structural results. The diagram is organisational: w is not the spin doublet, the face components are not additional spacetime dimensions, and the primitive scalar u is not the collective metric scalar.

The model developed in this paper therefore incorporates:

- a conservative microscopic action with explicit common and internal constitutive degrees of freedom;
- a C_3 -symmetric internal sector generating exactly three symmetry-related orientations;
- structured-branch Schur softening accompanied by correlated sixth-order regulation;
- a stable finite-wave-number scale-selection corridor;
- an explicit positive two-face occupation parent with conserved fixed Noether charge;
- a benchmark one-orbital localised matter branch with radial, non-radial and full-face stability controls;
- a relational family/flavour sector in which three C_3 -related orientations couple to the existing face/ u channel through the conjugate-domain response, with a stable benchmark nonlinear branch and no inserted family matrix or microscopic complex phase;

- a distinct covariantly attached phase frame carrying reduced spin-cover topology;
- positive-frequency Schrödinger-like evolution with the absolute elementary action scale left open;
- reduced exchange oddness and conditional mode CAR, separated from chirality and gauge structure;
- representative Born-form detector response without fundamental microscopic damping;
- a collective incompatible-strain construction reproducing the spatial Einstein/Ward structure;
- selected finite-truncation ultraviolet control, finite-core regulation, effective/core area-law scaling and a normalisation-independent cross-sector closure test.

Several frontiers therefore remain. The physical face coefficients must be derived from the microscopic action and tested blindly against the already-established fixed-charge localisation and stability corridor. The action-class family mechanism and its stable benchmark branch are now explicit, but the physical face/ u kernels and the two deeper constitutive cross-responses must still be descended independently before family masses or mixing matrices are calculated; the corresponding CKM and PMNS values therefore remain physical predictions rather than inputs. The universal elementary action scale must be selected without importing $\hbar$. The normalised $B = 1$ phase-frame construction is already available at the structural level, but its microscopic face/bundle descent and physical stiffness/pinning coefficients remain open. The reduced centre-odd spin structure and the independent exchange-odd representation are also already available at their stated levels; the complete microscopic theory must still select the corresponding elementary matter sector before a local relativistic fermion theory can be claimed. Local gauge dynamics, universal physical sampling of the already-identified invariant preparation-phase measure, the matter-to-gravity source projection, the closed-system common tensor response and nonlinear four-dimensional gravitational closure likewise remain open. The M1 relaxation length is the preferred microscopic ruler candidate, but its universal physical identification remains to be validated blindly. Strong-field metric regularity, horizon-state counting and full coupled ultraviolet completion are downstream problems rather than results assumed here.

Scope and level of claim. The present work should therefore be read as a microscopic reconstruction of the strongest defensible RSTM mechanisms, not as a claim that every sector of quantum field theory and gravitation has already been derived. Some results are analytic consequences of the admitted action; some are structural identities; and some are representative numerical existence or stability demonstrations. These levels are kept explicit throughout the paper and are consolidated in Supplementary Section P. In particular, the finite-wave-number corridor is a scale-selection result, the fixed-charge branch is an existence-class matter result, the family/flavour branch is an action-class numerical-control result with physical response descent still open, the spin and exchange constructions are reduced/topological results with a remaining physical-selection gate, the detector calculation is representative rather than a universal Born theorem, and the gravitational result is spatial/structural rather than completed four-dimensional GR.

The relationship to the earlier RSTM formulation is similarly selective. The 2-4-6 scalar hierarchy, positive-frequency envelope, two-face architecture, higher-gradient ultraviolet regulation, selected localisation mechanisms, effective flavour machinery and several phenomenological calculations remain useful. Their microscopic interpretation is revised where necessary. Fundamental damping, direct identification of scalar stiffness with Newton's constant, three primitive generation fields, automatic gauge completion, universal Born closure, all-scale Osterwalder-Schrader reconstruction and complete strong-field gravity are not carried forward as established microscopic results.

The aim is consequently narrower but more demanding: to determine how much familiar physics can arise from a small, conservative microscopic ontology without using the quantities to be explained as inputs. The calculations below show that this programme already produces a correlated finite-scale elastic response, a stable fixed-charge localised matter carrier, a reduced spin/exchange architecture, positive-frequency quantum-like evolution and a non-trivial collective spatial geometry. Whether the remaining physical normalisations and local field-theoretic structures descend from the same microscopic action provides a direct programme of further calculation and falsification.

Organisation of the Paper

- Section 2 (Methods) defines the microscopic variables and action, derives the Euler-Lagrange system, develops the structured-branch Schur reduction, the finite-scale analysis, the two-face occupation, relational family-response and phase-frame constructions, and the quantum, detector, gravitational and ultraviolet reduction methods.
- Section 3 (Results) presents the correlated stiffness response and finite-scale corridor, the on-shell parameter restriction, three internal orientations, recovery of the effective scalar law, fixed-charge matter localisation and stability, the stable relational family/flavour branch, positive-frequency quantum structure, exchange/CAR and chirality results, representative detector response, spatial Einstein/Ward construction, ultraviolet control, regulated-core behaviour and the cross-sector normalisation relation.
- Section 4 (Discussion) compares the common emergence mechanism across sectors, gives the continuity audit with the earlier RSTM formulation, distinguishes retained structures from superseded interpretations, and sets out the principal falsification tests and remaining open gates.
- Section 5 (Conclusion) summarises the microscopic reconstruction and the physical closures required before the framework can be regarded as a complete theory.

Supplementary Sections A-P provide the detailed supporting derivations, numerical audits, continuity analysis and claim-status definitions:

- **Supplementary Section A** - Full Microscopic Action and Euler-Lagrange System
- **Supplementary Section B** - Coercivity, Hamiltonian Positivity and Ghost Audit
- **Supplementary Section C** - Constitutive Relaxation and Schur-Complement Derivation
- **Supplementary Section D** - Finite-Scale Corridor and Wavelength Proof
- **Supplementary Section E** - On-Shell Microscopic Parameter Manifold
- **Supplementary Section F** - C3 Phase Structure, Relational Family Response and Flavour Controls
- **Supplementary Section G** - Fixed-Charge Matter Localisation and Stability
- **Supplementary Section H** - Positive-Frequency Reduction, Noether Action and Phase-Frame Construction

- **Supplementary Section I** - Exchange and CAR Derivation
- **Supplementary Section J** - Measurement, Threshold Competition, Effective Damping and Detector-Basin Model
- **Supplementary Section K** - Collective Metric, Incompatibility and Ward Derivation
- **Supplementary Section L** - Power Counting and Osterwalder-Schrader/Reflection-Positivity Audit
- **Supplementary Section M** - Regulated Cores and Area-Law Derivation
- **Supplementary Section N** - Cross-Sector Normalisation Invariant
- **Supplementary Section O** - Detailed Continuity Audit with the Earlier RSTM Formulation
- **Supplementary Section P** - Glossary, Notation and Claim-Status Key

2. Methods

The purpose of this section is to define the microscopic variables, conservative action class, reductions and diagnostic tests used in the calculations that follow. The construction is organised so that quantities appearing in a reduced description are obtained only after the relevant microscopic or collective degrees of freedom have been varied and, where appropriate, eliminated. Detailed coercivity proofs, stability calculations and numerical benchmarks are given in Supplementary Sections A-N; here we state the working equations and the order in which they are applied.

2.1 Microscopic ontology and conservative action

The microscopic core is formulated in $3 + 1$ dimensions. Its minimal primitive fields are a real scalar substrate coordinate $u(\mathbf{x}, t)$, a real common constitutive coordinate $s(\mathbf{x}, t)$, and a complex internal redistribution coordinate $w(\mathbf{x}, t)$:

$$u \in \mathbb{R}, \quad s \in \mathbb{R}, \quad w \in \mathbb{C}, \quad w = w_1 + iw_2.$$

The constitutive coordinate s is the common, or M1-bright, mode because it is permitted to enter the leading scalar bending response linearly. The two real components of w form an internal redistribution doublet. They obey the exact primitive constitutive symmetry

$$w \rightarrow e^{2\pi i/3} w.$$

The equivalent three-coordinate representation of the constitutive state is only a change of basis: it represents one common coordinate plus a two-real-dimensional relative sector and is not interpreted as three independent primitive family fields.

The closed microscopic action is decomposed as

$$S_{micro} = S_{core} + S_{face} + S_{int},$$

where S_{core} contains the substrate and constitutive variables, S_{face} contains the conservative two-face occupation sector, and S_{int} contains couplings generated within the closed action. A representative local core action is

$$S_{core} = \int dt d^3x [\mathcal{L}_u + \mathcal{L}_s + \mathcal{L}_w - V_c(s, w)],$$

with

$$\begin{aligned}\mathcal{L}_u &= \frac{\rho_u}{2} \dot{u}^2 - \frac{T_u}{2} |\nabla u|^2 - \frac{1}{2} K_4(s, |w|^2) (\Delta u)^2 - \frac{K_{6u}}{2} |\nabla \Delta u|^2 - U_u(u), \\ \mathcal{L}_s &= \frac{\rho_s}{2} \dot{s}^2 - \frac{Y_{s2}}{2} |\nabla s|^2 - \frac{Y_{s4}}{2} (\Delta s)^2 - \frac{Y_{s6}}{2} |\nabla \Delta s|^2, \\ \mathcal{L}_w &= \frac{\rho_w}{2} |\dot{w}|^2 - \frac{Y_{w2}}{2} |\nabla w|^2 - \frac{Y_{w4}}{2} |\Delta w|^2 - \frac{Y_{w6}}{2} |\nabla \Delta w|^2.\end{aligned}$$

The admitted microscopic existence class has positive kinetic coefficients and positive sixth-order principal terms, with $K_4(s, |w|^2) > 0$ throughout the unreduced microscopic domain. Time derivatives remain second order. Higher derivatives act spatially. These conditions allow the ultraviolet regulator to be retained without adding ordinary higher-time-derivative degrees of freedom. The complete microscopic action class is given in Supplementary Section A; the corresponding coercivity, Hamiltonian-positivity and ghost audit is given in Supplementary Section B.

No universal Rayleigh damping or unexplained external forcing is included in the closed microscopic action. Effective attenuation, memory and forcing may appear only after an environment, detector or other degrees of freedom have been eliminated or when an explicitly driven subsystem is selected. Likewise, the observed values of G , $\hbar$ and the Planck length are not inserted as microscopic coefficient definitions.

2.2 CPT-folded conjugate-domain architecture

The membrane is completed by two complementary face or conjugate-domain branches with reversed temporal orientation across the CPT fold. The two branches are treated as components of one conservative microscopic system rather than as independent positive- and negative-energy theories. Energy that appears with the opposite sign in a reduced one-face bookkeeping convention can therefore represent conservative transfer between complementary branches rather than an unbounded negative Hamiltonian of the full system. For comparison, CPT-symmetric cosmological constructions containing oppositely oriented pre- and post-Big-Bang branches have also been investigated [33]; the RSTM conjugate-domain architecture is independently formulated and is not derived from those models.

This architecture is used later in three distinct ways. First, it provides the face space on which the conservative occupation field is defined. Secondly, it supplies the complementary orientation required for the matter/antimatter and chirality discussion. Thirdly, it allows reduced one-face dynamics to appear non-conservative after the complementary sector has been omitted. These roles are kept separate: face reflection is not identified with fermion parity, two-particle exchange, or a completed local relativistic antiparticle field. The conjugate-domain interpretation, including its energy-sign and claim-boundary implications, is consolidated in Supplementary Section O.10. Temporal-mirror constructions are nevertheless useful as external mathematical precedent for treating a conjugate-domain junction as a matching problem rather than a pointwise identification [34]. In the present construction the two domain-resolved internal coordinates are therefore not rigidly slaved as $w_m = \overline{w_p}$ locally; such an identification would give $Y = \text{Im}(w_p w_m) = 0$ and remove the relational family response considered below. This comparison concerns matching methodology only and is not taken as the origin of the RSTM conjugate-domain architecture.

2.3 Euler--Lagrange dynamics

The equations of motion are obtained by varying the variable-coefficient action before any homogeneous or slowly varying approximation is imposed. For the minimal scalar truncation the result is

$$\rho_u \ddot{u} - T_u \Delta u + \Delta \left[K_4 \left(s, |w|^2 \right) \Delta u \right] - K_{6u} \Delta^3 u + U'_u(u) = J_u,$$

where J_u denotes the source generated by variation of $S_{face} + S_{int}$ with respect to u . The variable-coefficient fourth-order term is retained in divergence form. In particular, $\Delta [K_4 \Delta u]$ is not replaced by $K_4 \Delta^2 u$ when K_4 varies in space, because gradients of the local constitutive state then generate additional deterministic mode-scattering terms.

Variation with respect to the common constitutive coordinate gives

$$\rho_s \ddot{s} - Y_{s2} \Delta s + Y_{s4} \Delta^2 s - Y_{s6} \Delta^3 s + \frac{\partial V_c}{\partial s} + \frac{1}{2} \frac{\partial K_4}{\partial s} (\Delta u)^2 = J_s,$$

and the two real components $w_a = (w_1, w_2)$ obey the corresponding C_3 -covariant equations

$$\rho_w \ddot{w}_a - Y_{w2} \Delta w_a + Y_{w4} \Delta^2 w_a - Y_{w6} \Delta^3 w_a + \frac{\partial V_c}{\partial w_a} + \frac{1}{2} \frac{\partial K_4}{\partial w_a} (\Delta u)^2 = J_{w_a}.$$

At the symmetric point $w = 0$, C_3 invariance forbids an isolated scalar invariant linear in w , so the relative doublet is absent from the universal linear scalar constitutive response. The common coordinate s is therefore the leading constitutive channel used in the scalar Schur reduction. The complete variable-coefficient Euler-Lagrange variation and the resulting core equations are given in Supplementary Section A.

2.4 Constitutive relaxation and Schur reduction

The effect of a responsive but healthy microscopic coordinate is evaluated at quadratic order using a Schur complement. For a retained fluctuation ϕ and constitutive fluctuation χ , write

$$\delta^2 H = \frac{1}{2} \begin{pmatrix} \phi & \chi \end{pmatrix} \begin{pmatrix} A & C \\ C^T & H \end{pmatrix} \begin{pmatrix} \phi \\ \chi \end{pmatrix}.$$

Minimisation with respect to χ gives $\chi_{rel} = -H^{-1} C^T \phi$ and hence

$$A_{eff} = A - C H^{-1} C^T.$$

If H is positive, $C H^{-1} C^T$ is positive semidefinite. Relaxation can therefore soften the retained response without requiring a negative microscopic kinetic or potential direction.

The background about which this reduction is performed is essential. Around a homogeneous scalar vacuum with $\Delta \bar{u} = 0$, the leading interaction $\sigma(\Delta \phi)^2$ is cubic in fluctuations and gives no tree-level quadratic two-point correction. On a curvature-populated branch with $Q_0 \equiv \Delta \bar{u} \neq 0$, the same microscopic coupling generates the bilinear term $\Gamma_s \sigma \Delta \phi$, where $\Gamma_s = K_{4,s} Q_0$. In static Fourier space the common constitutive kernel is written

$$H_s(k) = D_{s0} + Y_s k^2 + Y_{s4} k^4 + Y_{s6} k^6 + \dots,$$

with $D_{s0} > 0$. Eliminating σ gives the exact relaxed scalar kernel

$$A_{eff}(k) = A_u(k) - \frac{\Gamma_s^2 k^4}{H_s(k)}.$$

For $k^2 \uparrow_s^2 \ll 1$, where $\uparrow_s^2 = Y_s / D_{s0}$, expansion of H_s^{-1} yields

$$\Delta K_4 = -\frac{\Gamma_s^2}{D_{s0}} < 0, \quad \Delta K_6 = \frac{\Gamma_s^2 Y_s}{D_{s0}^2} > 0,$$

and therefore

$$\Delta K_6 = -\uparrow_s^2 \Delta K_4.$$

The exact rational kernel, rather than the sign of an individual coefficient in its derivative expansion, is used for the stability audit. At finite frequency the same elimination generates a frequency-dependent self-energy and hence a non-local memory kernel in the u -only description [35,36]. The full structured-background Hessian reduction, exact rational kernel and derivative expansion are derived in Supplementary Section C.

2.5 Dimensionless finite-scale analysis

The reduced scalar gradient kernel is analysed after constitutive relaxation. For a homogeneous effective state,

$$P(k) = Tk^2 + K_{4,eff}k^4 + K_{6,eff}k^6,$$

with $T > 0$ and $K_{6,eff} > 0$. The finite-scale branch of interest has $K_{4,eff} < 0$. Introduce the sixth-order crossover length and dimensionless variables

$$\downarrow_6 = \left(\frac{K_{6,eff}}{T} \right)^{1/4}, \quad q = k\downarrow_6, \quad \beta_c = \frac{K_{4,eff}}{\sqrt{TK_{6,eff}}}.$$

Then

$$P(k) = \frac{T^{3/2}}{K_{6,eff}^{1/2}} (q^2 + \beta_c q^4 + q^6).$$

Global positivity and the appearance of a distinct non-zero stationary maximum/minimum pair are tested separately. For this local 2 – 4 – 6 effective kernel, the stable finite-wave-number corridor is

$$-2 < \beta_c < -\sqrt{3},$$

equivalently

$$3TK_{6,eff} < K_{4,eff}^2 < 4TK_{6,eff}, \quad K_{4,eff} < 0.$$

Inside this corridor the preferred minimum is at

$$q_*^2 = \frac{-\beta_c + \sqrt{\beta_c^2 - 3}}{3}, \quad k_* = \frac{q_*}{\downarrow_6},$$

with wavelength $\lambda_* = 2\pi/k_*$. The two corridor boundaries are retained as distinct diagnostics: $|\beta_c| = \sqrt{3}$ marks the first appearance of the stationary pair, whereas $|\beta_c| = 2$ is the marginality boundary at which the finite- k minimum reaches zero. The complete corridor derivation, boundary analysis and preferred-wavelength bounds are given in Supplementary Section D.

2.6 On-shell microscopic parameter reduction

The quantities entering the effective kernel are not treated as independently adjustable microscopic constants. On the symmetric internal branch $w = 0$, take a slowly varying local background with $Q = \Delta\bar{u}$. The s -dependent static energy density is

$$\mathcal{E}_{loc} = V_s(s) + \frac{1}{2}K_4(s)Q^2 + \dots$$

Constitutive equilibrium requires

$$V'_s(s) + \frac{1}{2}K'_4(s)Q^2 = 0,$$

so that, when $K'_4(s) \neq 0$,

$$Q^2 = -\frac{2V'_s(s)}{K'_4(s)}.$$

The local common-mode Hessian and scalar-constitutive mixing are then

$$D_s = V''_s(s) + \frac{1}{2}K''_4(s)Q^2, \quad \Gamma_s = K'_4(s)Q.$$

Thus Q , D_s and Γ_s are functions of the same on-shell constitutive branch. The corresponding $K_{4,eff}$ and $K_{6,eff}$ obtained by Schur reduction occupy an action-derived manifold in effective-coefficient space. The finite- k corridor is applied to that on-shell manifold rather than to an unconstrained phenomenological parameter plane. The equilibrium reduction and resulting on-shell effective-coefficient manifold are developed in Supplementary Section E.

2.7 Minimal internal sector and three orientations

The internal redistribution sector is analysed using the lowest-order bounded C_3 -invariant potential

$$V_c(s, w) = V_s(s) + \frac{1}{2}M_w^2(s)|w|^2 - \kappa_3 \text{Re}(w^3) + \frac{\lambda_w}{4}|w|^4, \quad \lambda_w > 0.$$

Writing $w = re^{i\theta}$ gives

$$V_c = V_s(s) + \frac{1}{2}M^2r^2 - \kappa_3r^3 \cos(3\theta) + \frac{\lambda_w}{4}r^4.$$

For $\kappa_3 > 0$ the angular minima satisfy $\cos(3\theta) = 1$, so

$$\theta_n = \frac{2\pi n}{3}, \quad n = 0, 1, 2.$$

The opposite sign of κ_3 shifts all three angles by $\pi/3$ without changing their multiplicity. Radial stationary points and the associated first-order transition are obtained from $\partial V/\partial r = 0$, with radial and angular Hessians used to test local stability. The important methodological separation is that C_3 is the primitive microscopic postulate, whereas the three symmetry-related orientations are derived from the allowed anisotropy. The radial transition, positive radial and angular gaps, and three-orientation phase structure are derived in Supplementary Section F.

2.8 Recovery of the effective scalar hierarchy

After the constitutive variables have been solved, relaxed or integrated out, the scalar sector is described by an effective action $\Gamma_{eff}[u]$. Where a local derivative expansion is valid, its leading equation is

$$\rho_{eff}\ddot{u} - T_{eff}\Delta u + K_{4,eff}\Delta^2 u - K_{6,eff}\Delta^3 u + V_{eff,u} = 0.$$

The familiar 2--4--6 RSTM hierarchy is therefore retained as a collective law. Its coefficients are correlated outputs of the microscopic response, and the exact reduced theory may be non-local in space and time outside the derivative-expansion regime. Damping, stochastic-looking forcing or externally prescribed sources are added only for explicitly reduced, driven or open-system branches and are not reintroduced into the closed microscopic equation. The detailed action-to-effective-scalar descent and its parameter correlations are distributed across Supplementary Sections A-E.

2.9 Face occupation, fixed charge and phase-frame construction

2.9.1 Conservative face occupation sector

Persistent matter localisation is formulated in a separate two-face occupation sector. The local-frame occupation coordinate is $\psi(\mathbf{x}, t) \in \mathbb{C}^2$, representing four real canonical configuration variables after realification. A representative action is

$$S_{occ} = \frac{1}{2} \int dt d^3x \left[\rho_F \dot{\psi}^\dagger \dot{\psi} - \psi^\dagger M_F^2(s, u) \psi - z_2 \partial_i \psi^\dagger \partial_i \psi - z_4 \Delta \psi^\dagger \Delta \psi - z_6 \partial_i \Delta \psi^\dagger \partial_i \Delta \psi \right],$$

with $\rho_F > 0$, $z_2 > 0$, $z_4 \geq 0$ and $z_6 > 0$. The face mass operator is selected from the positive-square class

$$M_F^2(s, u) = m_f^2 I + P^2(s, u),$$

where P is Hermitian. Consequently $M_F^2 \geq m_f^2 I$ globally. The lower-face response used in the fixed-charge branch is generated from the same positive square that also produces the odd inter-face transition and its positive even completion; these pieces are not introduced as independent stabilising repairs. The complete positive-square face parent is constructed in Supplementary Section A, with its global coercivity and dormant-face positivity conditions audited in Supplementary Section B.

2.9.2 Exact common phase symmetry and Noether charge

The face action is restricted to Hermitian bilinears and is postulated to possess the exact common phase symmetry

$$\psi \rightarrow e^{i\alpha} \psi.$$

After realification this is an ordinary internal $O(2)$ rotation. Noether's theorem gives [10]

$$Q_F = \rho_F \int d^3x \operatorname{Im}(\psi^\dagger \dot{\psi}).$$

For a normalised circular mode $\psi = A\phi(\mathbf{x}) e^{-i\omega t}$, the charge satisfies $Q_F = \omega A^2$ in the adopted canonical normalisation. The charge is held fixed in the localisation problem. This fixed-charge variational setting is analogous in general spirit to the conserved-charge localisation principle of Q-ball theory [11]. Conserved Noether charge may also coexist with non-trivial topology, as in Q-lump constructions [12]. Neither fact is therefore treated as novel here. The RSTM distinction lies instead in the positive-square two-face parent and reciprocal coupling to the common constitutive coordinate that generates the occupied-face mass well, together with the separation of that binding mechanism from the phase-frame topology introduced below. The exact phase symmetry and Noether charge are derived in Supplementary Section A; their fixed-charge matter interpretation is developed in Supplementary Section G.

2.9.3 Reciprocal fixed-charge localisation

On the occupied lower face, the circular ansatz reduces the matter problem to a static reciprocal pair. The spatial occupation mode obeys

$$\left[m_-^2(s) + Z_2(-\Delta) + Z_4(-\Delta)^2 + Z_6(-\Delta)^3 \right] \phi = \omega^2 \phi$$

while the common constitutive coordinate satisfies the fixed-charge equation

$$H_s s + \frac{Q_F}{2\omega} (\kappa_0 + \chi_s s) \phi^2 = 0$$

in the benchmark truncation. The reduced fixed-charge energy is $E_{Q_F} = E_s + Q_F\omega$. Localisation is assessed against the dispersed continuum carrying the same charge,

$$E_{Q_F} < m_\infty Q_F.$$

The one-particle candidate is required to contain one preferred spatial bound orbital. Radial evolution, non-radial angular sectors and the dormant complementary-face block are then tested separately for stability rather than inferred from the radial existence calculation alone. The complete one-orbital fixed-charge construction, continuum comparison and radial, non-radial and full-face stability tests are given in Supplementary Section G.

2.9.4 Relational conjugate-domain family response

Exact C_3 symmetry generates three internal orientations but leaves every universal C_3 -invariant scalar response equal on the three members of the orbit. Physical differentiation must therefore arise downstream rather than from an explicitly inserted family vector. Resolve the redistribution field on the two conjugate domains as w_p and w_m , denoting the plus and minus conjugate-domain coordinates, and define

$$Y \equiv \text{Im}(w_p w_m)$$

Under the physical single-fold exchange, Y is odd. The conjugate reference remains independently responsive; rigid local identification $w_m = \overline{w_p}$ is excluded because it makes Y vanish identically. The minimal positive-square face completion is

$$P_{61} = a(s)I + b(s)\tau_1 + [c_u u + c_Y Y] \tau_2$$

with

$$M_F^2 = m_f^2 I + P_{61}^2$$

The coefficient c_Y is a single real, generation-universal face-domain response. No microscopic complex coupling is inserted: the relative imaginary structure required for a CP-capable transition is supplied by the fixed Hermitian face matrix, $\langle +|\tau_2|-\rangle = -i$. The even radial response of the internal sector is linked to the common constitutive state by $\lambda_{sw}s|w|^2$. These two couplings perform different functions: the fold-odd response distinguishes the family-relative channel, whereas the $s - w$ response transmits its radial dressing into the common matter environment.

2.9.5 Odd-sector branch selection and response invariants

The flavour-sensitive instability is treated as a coupled response problem, not as a fitted three-family matrix. In the minimal reduction the odd Hessian is

$$H_{\text{odd}} = \begin{pmatrix} D_\eta & G & J \\ G^\dagger & H_u & 0 \\ J^\dagger & 0 & H_\delta \end{pmatrix}$$

where the first coordinate is the τ_2 -carrying face quadrature, u is the existing microscopic odd constitutive coordinate and δ is the local relative-angle response of w_p and w_m . Eliminating the stable gapped channels gives

$$K_{\text{odd}}^{\text{eff}} = D_\eta - GH_u^{-1}G^\dagger - JH_\delta^{-1}J^\dagger$$

The physical onset criterion is the zero crossing of the lowest constrained eigenvalue of this operator after removal of the common-phase and translational symmetry directions. Field-normalisation dependence can be removed from the two deeper constitutive bridges. For the common/radial response,

$$\epsilon_{sw} = \frac{H_{sr}}{\sqrt{H_{ss}H_{rr}}} = -\frac{\chi_{sr}}{\sqrt{\chi_{ss}\chi_{rr}}}$$

while for the fold-face response,

$$\rho_{fold} = -\frac{\chi_{2\delta}}{\sqrt{\chi_{22}\chi_{\delta\delta}}}$$

with the equivalent stripped family susceptibility

$$\Pi_Y = c_Y^2 R H_\delta^{-1} R$$

These are microscopic response targets rather than flavour-fit parameters. Quantities such as the face splitting, u-kernel and particle-profile responses are to be descended directly from the existing action. Neither observed fermion masses nor CKM/PMNS quantities enter the branch-selection calculation.

2.9.6 Normalised phase-frame bundle

The local occupation amplitude and the topological phase frame are treated as distinct collective variables. A normalised frame $U_F(\mathbf{x}, t) \in SU(2)$ is introduced independently of the asymptotically vanishing occupation modulus, and the physical face state is reconstructed as

$$\Upsilon(\mathbf{x}, t) = U_F(\mathbf{x}, t) \psi(\mathbf{x}, t).$$

The moving-frame connection is

$$\Gamma_\mu = -(\partial_\mu U_F) U_F^\dagger, \quad D_\mu = \partial_\mu + \Gamma_\mu,$$

so that

$$D_\mu(U_F \psi) = U_F \partial_\mu \psi.$$

Pure orientation is therefore isospectral to the texture-free occupation operator. Binding remains in the amplitude/mass sector while U_F carries the topology and spin transport. The frame is supplied with its own positive second-, fourth- and sixth-order spatial functional. At the level considered here this frame functional is spatial only; no independent frame kinetic term or propagating vacuum frame degree of freedom is assumed. Temporal spin-state evolution remains carried by the positive-frequency matter sector.. The microscopic descent of U_F from the deeper primitive u, s, w substrate is not assumed and remains a separate closure problem. The normalised frame bundle, covariant isospectral attachment, finite frame-scale analysis and reduced spin construction are developed in Supplementary Section H.

2.10 Positive-frequency quantum construction

The complex structure used in the reduced quantum description is obtained from stable real canonical modes. For a normal mode with $\omega_n > 0$, reality relates the positive- and negative-frequency pieces. Restricting to the positive-frequency branch defines the canonical quarter-turn J on real phase space, with $J^2 = -I$, and hence a compatible complex structure.

The use of a selected oscillatory carrier branch to obtain first-order complex evolution from the real second-order STM dynamics predates the present microscopic reconstruction. In RSTM v3, posted on 28 May 2025, the multiple-scale ansatz $u = U(X, T)e^{i(k_0 x - \omega_0 t)}$, together with $\partial_t \rightarrow -i\omega_0 + \varepsilon^2 \partial_T$, generated the $2i\rho\omega_0 \partial_T U$ term from which the first-order Schrödinger-like envelope equation followed [37]. That earlier construction selected a single $e^{-i\omega_0 t}$ carrier branch in practice, but did not formulate the reduction as a general projection theorem on the positive-frequency solution subspace. A later independent treatment by Singh gives that latter step

an explicit mathematical formulation using positive-frequency selection, the temporal Hilbert transform and the corresponding Hardy-space structure [38]. The present RSTM construction therefore retains continuity with its earlier carrier-branch reduction while placing it in a cleaner canonical positive-frequency framework. Its additional RSTM-specific content is the application of that reduction to the already-selected fixed-charge face carrier and the carrier-specific action identification developed below.

For the two canonical matter modes (Q_a, P_a) , an action normalisation J_* defines

$$\psi_a = \frac{Q_a + iP_a}{\sqrt{2J_*}}, \quad a = 1, 2.$$

A positive-frequency state then evolves according to

$$iJ_*\partial_t\psi = H_{eff}\psi.$$

On the exact circular face carrier, the spectral, symplectic and phase-generator definitions of the classical action coincide with the fixed face charge,

$$J_{phase} = J_{spectral} = J_{symplectic} = Q_F.$$

This identifies the carrier action structurally, but it does not determine the universal dimensional value J_* . The classical equations are invariant under a global rescaling of the complete action, whereas momenta, energies and action variables rescale. $J_* = \hbar$ is therefore reserved for a downstream blind comparison rather than imposed as a microscopic definition. The positive-frequency complex structure, equality of the carrier-action definitions and the remaining absolute-normalisation problem are treated in Supplementary Section H.

2.11 Spin, chirality and exchange as distinct structures

2.11.1 Spin-cover structure

The reduced one-particle sector contains two independent positive-frequency canonical modes and therefore a rank-two complex fibre $E_{spin} \simeq \mathbb{C}^2$. Transport with the degree-one $SU(2)$ phase frame gives the double-cover holonomy

$$W(2\pi) = -I_2.$$

This establishes the reduced one-particle rotational character at the stated topological level. It is not used as a substitute for the independent two-particle exchange calculation. The complete reduced spin-cover construction and its positive-residue audit are given in Supplementary Section H.

2.11.2 Face/orientation chirality

Chirality is introduced through the two-face/orientation sector rather than through the phase-frame rotation sign. At effective level the projector architecture is written schematically as

$$P_W = F_+P_L + F_-P_R,$$

which is compatible with left-handed visible-face currents and the chirality-reversed complementary construction. This is a structural chiral architecture, not yet a completed local chiral gauge theory. The face/orientation chiral precursor and its stated limitations are developed in Supplementary Section H.

2.11.3 Exchange sector

Two-particle statistics are studied on a global degree-two texture whose closed path exchanges two degree-one constituents. The local one-core spaces are followed using a moving two-patch atlas and the relative exchange holonomy is extracted only after single-particle geometric phases are divided out. Conditional on physical selection of the resulting odd sector, the reduced one-particle space is antisymmetrised by exterior-Fock construction and the standard mode canonical anticommutation relations follow. The microscopic compulsory selector of that odd sector and the complete local relativistic CAR field algebra remain outside the present Methods closure. The independent two-particle exchange geometry, antisymmetric exterior-Fock construction and conditional mode CAR are given in Supplementary Section I.

2.12 Internal-frame covariance and charge-class constraints

Local changes of internal oscillator frame are represented by a connection so that observables are independent of the chosen local internal basis. This supplies the structural gauge-connection architecture. Conditional on faithful action of the oscillator/internal frame, the globally acting bundle is constrained to

$$G_{faith}^{RSTM} = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\mathbb{Z}_6},$$

with the corresponding charge-class congruence

$$Y \equiv \frac{\sigma}{2} - \frac{\tau}{3} \pmod{1}.$$

The $\mathbb{Z}_6$ quotient is a recognised possible global form of the Standard Model gauge group [39]; here its appearance is conditional on the admitted RSTM oscillator-frame action. The global bundle and charge classes are therefore more constrained than an arbitrary direct product of matrices. However, local physical gauge redundancy, the Yang--Mills kinetic terms [40] and the physical couplings g_1 , g_2 and g_3 are not assumed to have been derived by this construction. The structural gauge-connection and global charge-class result, together with its claim boundaries, is consolidated in Supplementary Section O.5.

2.13 Open-system reduction and measurement

2.13.1 Conservative analyser and Born-form channel weights

A positive-frequency state $|\psi\rangle$ is presented to detector channels $|d_i\rangle$. The linear analyser amplitude is

$$A_i = \langle d_i | \psi \rangle.$$

If absorbed detector energy is quadratic in the incident amplitude, normalised channel weights are

$$P_i^{(wt)} = \frac{|\langle d_i | \psi \rangle|^2}{\sum_j |\langle d_j | \psi \rangle|^2},$$

or, for a general positive detector operator E_i ,

$$P_i^{(wt)} = \langle \psi | E_i | \psi \rangle, \quad \sum_i E_i = I.$$

This step derives the quadratic channel-response form [17]; it does not assume that arbitrary physical preparations already sample those weights with the corresponding outcome frequencies.

2.13.2 Deterministic threshold competition and detector basins

Macroscopic recording is modelled as deterministic competition between detector channels. Each channel integrates its local drive until a nonlinear threshold is crossed; the winning branch amplifies and relaxes into a persistent detector record. The complete state evolves deterministically, $X_{micro}(t) = \Phi_t[X_{micro}(0)]$, but a macroscopic preparation need not specify every microscopic detector or environmental variable λ . The outcome map $D_\psi : \lambda \mapsto i$ defines basins $B_i(\psi)$, and a preparation measure μ_{prep} would give

$$P_i = \int_{B_i(\psi)} d\mu_{prep}(\lambda).$$

Threshold-detector schemes in which classical or random signals generate Born-compatible click statistics have established precedent [41]. The present construction therefore does not treat threshold detection itself as novel. Its narrower purpose is to place deterministic threshold competition downstream of the conservative RSTM analyser and to test whether unresolved detector dynamics can sample the already-established quadratic channel weights while producing a persistent macroscopic record.

The representative calculation reported below uses a reduced deterministic Lorenz/SRB detector proxy to sample unresolved apparatus states; this proxy is not identified with the fundamental conservative RSTM substrate dynamics. Its complete implementation and scanned control parameters are supplied in Supplementary Code S3.

The universal Born problem is therefore reduced to a definite measure question: whether the physically generated preparation measure gives basin measures equal to the quadratic channel weights for arbitrary measurement contexts.

The unresolved preparation variables should not be identified with the coherent collective relations that define the prepared matter state. In particular, a fixed preparation may preserve the relative phase and orientation of the reduced rank-two spin fibre while leaving finer microscopic carrier, detector and environmental variables unspecified. Statistical variation of those unresolved variables is therefore compatible with coherent spin preparation and does not, by itself, imply randomisation of the phase-frame structure.

2.13.3 Effective damping and memory

Irreversibility is introduced only after reduction. The combined field-detector-environment Hamiltonian remains conservative, while elimination of unobserved variables may generate a causal memory kernel [35,36] or, under a short-memory approximation, a local damping term in the reduced detector equation. Damping is used to stabilise records and describe apparatus energy flow; it is not used to generate the analyser amplitudes or their squared-overlap weights. The analyser, threshold-competition, detector-basin and preparation-measure analysis is given in Supplementary Section J.

2.14 Collective strain and spatial metric construction

The gravitational sector is constructed from collective moments of the microscopic state rather than by identifying the primitive scalar u directly with the gravitational metric. A collective scalar ϕ and traceless quadrupole Q_{ab} are soldered into the spatial metric by a local reference frame Θ_i^a ,

$$\gamma_{ij} = \updownarrow^2 \Theta_i^a \left[e^{2\phi I + Q} \right]_{ab} \Theta_j^b.$$

At linear order,

$$h_{ij} = 2Q_{ij} + 2\phi\delta_{ij}.$$

An ordinary displacement ξ_i produces the compatible strain

$$\varepsilon_{ij}^{comp} = \frac{1}{2} (\partial_i \xi_j + \partial_j \xi_i), \quad Inc(\varepsilon^{comp})_{ij} = 0,$$

and is interpreted as a reference or ruler deformation. Physical spatial curvature is instead associated with an incompatible collective strain ε_{ij} for which $Inc(\varepsilon)_{ij} \neq 0$. With $h_{ij} = 2\varepsilon_{ij}$,

$$G_{ij}^{(1)}[h = 2\varepsilon] = Inc(\varepsilon)_{ij}.$$

The collective decomposition admits the spatial relabelling

$$\delta\phi = \frac{1}{3}\partial_i \xi_i, \quad \delta Q_{ij} = \partial_{(i} \xi_{j)} - \frac{1}{3}\delta_{ij}\partial_k \xi_k,$$

which gives $\delta h_{ij} = 2\partial_{(i} \xi_{j)}$. Requiring the compatible directions to be null directions of the leading parity-even two-derivative kernel fixes its coefficient ratio to

$$a : b : c : d = 1 : -2 : 2 : -2.$$

The individual ingredients of this construction have relevant precedents. The linear relation between Saint-Venant incompatibility and the spatial Einstein tensor is established in continuum elasticity [18], while gauge invariance of a massless spin-two field is known to constrain the linear two-derivative dynamics to the Einstein/Fierz-Pauli structure [19]. More recently, an intrinsic D_4 Regge-lattice construction has combined closely related ideas: compatible vertex displacements form the diffeomorphism kernel, incompatible intrinsic edge-length modes carry curvature, and the resulting lattice kinetic operator reproduces the linearised Einstein/Fierz-Pauli operator [32]. The RSTM-specific result is therefore narrower. The collective gravitational variable is first assembled from microscopic RSTM state moments rather than postulated as an intrinsic metric or edge-length field; compatible collective deformation is then identified with spatial-reference redundancy; and requiring those reference directions to be null directions of the most general admitted collective two-derivative kernel fixes its coefficient ratio to the Einstein value.

This Ward-selected spatial Einstein structure is the target collective gravitational operator. The physical tensor pole, common propagation cone, absolute gravitational normalisation, temporal Fierz--Pauli recombination [19] and nonlinear hypersurface-deformation algebra [20] are deliberately left for downstream closure. The compatible/incompatible strain construction, spatial Einstein identity and Ward-coefficient derivation are given in Supplementary Section K.

2.15 UV, core and normalisation methods

2.15.1 Weighted ultraviolet counting

The microscopic fields remain second order in time and contain spatial derivatives through sixth order. At sufficiently large k the scalar dispersion behaves as

$$\omega^2 \simeq \frac{K_{6u}}{\rho_u} k^6, \quad \omega \propto k^3,$$

so the selected microscopic truncation is analysed using anisotropic scaling $\mathbf{x} \rightarrow b\mathbf{x}$, $t \rightarrow b^3 t$ with $z = 3$. In $d = 3$ the real scalar and constitutive fields have zero weighted dimension. The sixth-order principal terms are marginal while the displayed lower-derivative and local interaction vertices have positive weighted coupling dimension. Power counting is therefore applied only to the declared finite u, s, w truncation; $z = 3$ is not used to claim a unique finite exact functional theory.

For the free stable scalar covariance, Euclidean continuation is tested for reflection positivity with respect to the preferred Euclidean time. This is kept distinct from a full Lorentz-invariant

Osterwalder--Schrader reconstruction of the complete interacting theory. The weighted power-counting audit and free time-reflection-positivity analysis are given in Supplementary Section L.

2.15.2 Finite-core scaling and area-law diagnostic

The sixth-order energy of a localised scalar profile with characteristic amplitude U and radius R scales as

$$E_6 \sim \frac{K_6 U^2}{R^3}.$$

For $K_6 > 0$ this diverges as $R \rightarrow 0$ at fixed non-zero amplitude and therefore supplies an energetic obstruction to zero-width collapse. A finite physical core nevertheless requires competition with the remaining binding and nonlinear terms. Microscopic/core regularity is not identified directly with regularity of the emergent spacetime metric.

The retained core entropy calculation is evaluated separately at its effective level, where the leading term scales with the bounding area. It is not used as an input to the gravitational normalisation or to an independent microscopic horizon-state count. The finite-core scaling, regularity distinctions and effective/core area-law calculation are given in Supplementary Section M.

2.15.3 Cross-sector normalisation

The cross-sector closure uses three quantities whose physical values must be determined independently: the microscopic crossover/ruler scale $\downarrow_*$, the elementary action scale J_* , and the collective gravitational coupling G_{RSTM} . Their physical values are not inferred from one another. Their natural dimensionless comparison is

$$C_* = \frac{c^3 \downarrow_*^2}{J_* G_{RSTM}}.$$

Using the metric-normalised tensor stiffness Z_ε , with $G_{RSTM} = c^4 / (8\pi Z_\varepsilon)$, and defining the corresponding dimensionless stiffness ζ_ε and action scale $\hat{J}_*$, the same relation becomes

$$C_* = \frac{8\pi \zeta_\varepsilon}{\hat{J}_*}.$$

Because a global multiplication of the classical action rescales J_* and the inverse gravitational coupling in compensating ways, this combination is independent of the arbitrary overall action normalisation. The physical values of $\downarrow_*$, J_* and G_{RSTM} are therefore to be obtained independently and compared only downstream with $\downarrow_P$, $\hbar$ and G . The independent-scale definitions, action-rescaling audit and normalisation-independent closure relation are derived in Supplementary Section N.

These methods define the reduction chain used in Section 3. The principal tests are consequently not independent coefficient fits: they are whether one conservative microscopic action admits a healthy structured branch, lands on the finite-scale corridor, supports the fixed-charge matter branch and its stability conditions, supplies the reduced quantum and topological structures, and yields the required collective gravitational and cross-sector normalisations without importing the observed quantities that those calculations are intended to explain.

2.16 Computational reproducibility and supplementary code

Computational checks supporting the principal analytic and numerical claims are supplied as Supplementary Code S1-S5. S1 reproduces the local 2 – 4 – 6 finite-scale corridor, preferred wave

number and wavelength bounds of Supplementary Section D and independently checks exact-rational-kernel consistency for the explicit microscopic existence class of Supplementary Section C.11.1; S2 contains the recovered fixed-Noether-charge continuation, direct radial evolution, non-radial and translation-mode stability tests, charge-window robustness checks and final positive-square complementary-face audit of Supplementary Section G; S3 contains the deterministic SRB/Lorenz threshold-detector calculation, finite control-parameter scan, bootstrap analysis and negative control of Supplementary Section J; S4 symbolically verifies the Einstein/incompatibility identity and Ward coefficient ratio of Supplementary Section K; and S5 contains the family/flavour action-class closure programme, including the normalised family/flavour action-class control, radial onset, nonlinear Lyapunov-Schmidt calculation, full fixed- Q_F flavour-sensitive continuation, higher-angular and co-rotating stability tests, response-invariant identities, field-rescaling checks and fold-response provenance/no-go controls. The supplementary package also records the numerical grids, boundary and normalisation conventions, continuation and eigensolver settings and reference outputs used for the reported benchmarks. These calculations support existence, stability or structural diagnostics at the claim levels stated in the paper; benchmark coefficients are not physical parameter predictions.

3. Results

The results are organised around six principal outputs of the microscopic reconstruction — Structure, Scale, Particle, Family, Quantum and Geometry — followed by ultraviolet, core and cross-sector consistency tests.

Structure and Scale are developed first: healthy constitutive relaxation generates correlated intermediate-gradient softening and sixth-order regulation, restricts the effective coefficients to an on-shell microscopic manifold and produces the finite-wave-number corridor. Particle then establishes that scale selection and matter binding are distinct problems, with persistent localisation supplied instead by the conservative fixed-charge face sector. Family adds the relational conjugate-domain response and tests the resulting flavour-sensitive fixed-charge branch. Quantum collects the positive-frequency, spin/exchange and detector reductions that arise downstream of that matter construction. Geometry gives the incompatible-strain Einstein identity and Ward-selected spatial curvature structure. The final results test ultraviolet behaviour, finite-core regulation and the normalisation-independent cross-sector relation.

These results are not all asserted at the same epistemic level. Analytic derivations, structural identities, reduced/topological results and representative numerical existence demonstrations are distinguished explicitly throughout and consolidated in Supplementary Section P.

In section-number terms, Sections 3.1-3.6 concern the emergence of the effective scalar structure and its preferred finite scale; Section 3.7 establishes the explicit fixed-charge matter branch; Section 3.8 establishes the relational family/flavour branch; Sections 3.9-3.13 develop the reduced quantum, exchange, gauge, chirality and measurement structures; Sections 3.14-3.15 give the collective spatial gravity result; and Sections 3.16-3.18 close with ultraviolet, core and cross-sector consistency. Throughout, benchmark numerical values are used only to establish existence, stability or representative behaviour unless a result is explicitly stated to be parameter-independent.

3.1 Negative intermediate-gradient stiffness emerges from healthy microphysics

On a curvature-populated structured branch, relaxation of the healthy common constitutive coordinate produces the Schur correction

$$\Delta K_4 = -\frac{\Gamma_s^2}{D_{s0}} < 0.$$

The sign does not originate from a negative microscopic bending coefficient. The unreduced parent theory retains positive kinetic energy and positive microscopic fourth- and sixth-order

principal responses; the negative contribution appears only after the positive constitutive sector has been relaxed. The result is therefore an ordinary lowering of the retained energy through a Schur complement [2] rather than the introduction of a microscopic unstable or ghost-like mode. The background distinction is essential. Around a homogeneous scalar vacuum with $\Delta\bar{u} = 0$, the leading modulation vertex remains cubic and gives no tree-level quadratic two-point correction. On a structured branch with $Q_0 \equiv \Delta\bar{u} \neq 0$, the same coupling becomes an off-diagonal Hessian term, $\Gamma_s = K_{4,s}Q_0$, and the relaxed quadratic response is non-zero. The full structured-background derivation is given in **Supplementary Section C**.

3.2 Sixth-order regulation is generated simultaneously

The same constitutive response that lowers the relaxed fourth-order stiffness raises the sixth-order coefficient. Expanding the exact constitutive kernel at wavelengths long compared with its response length gives

$$\Delta K_6 = \frac{\Gamma_s^2 Y_s}{D_{s0}^2} > 0,$$

with

$$\Delta K_6 = -\downarrow_s^2 \Delta K_4, \quad \downarrow_s^2 = \frac{Y_s}{D_{s0}}.$$

Intermediate-gradient softening and short-wavelength regulation are therefore correlated consequences of one healthy microscopic relaxation channel. The effective fourth- and sixth-order coefficients do not move independently: increasing the magnitude of the Schur softening automatically increases the associated sixth-order stabilisation. This is the compact **structure from stability, not instability** result of the scalar sector. The exact rational kernel and derivative expansion are derived in **Supplementary Section C**.

3.3 A stable finite-scale corridor selects a preferred wavelength

For the reduced gradient kernel

$$P(k) = Tk^2 + K_{4,eff}k^4 + K_{6,eff}k^6,$$

with $T > 0$, $K_{6,eff} > 0$ and $K_{4,eff} < 0$, define

$$\downarrow_6 = \left(\frac{K_{6,eff}}{T} \right)^{1/4}, \quad q = k\downarrow_6, \quad \beta_c = \frac{K_{4,eff}}{\sqrt{TK_{6,eff}}}.$$

Combining global positivity with the requirement for a distinct non-zero stationary maximum/minimum pair gives the open corridor

$$-2 < \beta_c < -\sqrt{3},$$

or equivalently

$$3TK_{6,eff} < K_{4,eff}^2 < 4TK_{6,eff}, \quad K_{4,eff} < 0.$$

The two boundaries are physically different. At $\beta_c = -\sqrt{3}$ the finite- k stationary pair first appears; at $\beta_c = -2$ the preferred finite- k minimum becomes marginal and the positive-gradient stability bound is saturated. Within the corridor the preferred mode is

$$q_*^2 = \frac{-\beta_c + \sqrt{\beta_c^2 - 3}}{3}, \quad k_* = \frac{q_*}{\downarrow_6}.$$

Within the local $2 - 4 - 6$ effective kernel, the corresponding wavelength obeys the parameter-independent interval

$$2\pi < \frac{\lambda_*}{\downarrow 6} < 2\pi 3^{1/4},$$

or numerically

$$6.28 < \frac{\lambda_*}{\downarrow 6} < 8.27.$$

For the explicit microscopic existence class underlying this reduction, direct analysis of the unexpanded rational Schur kernel confirms that the positive finite- k minimum persists throughout the admitted witness interval. In that class $k_{*,ex} \uparrow_s < 0.153$; the exact preferred wave number differs from the sixth-order derivative-expansion result by less than 2.46% even at the limiting corridor boundary and by below approximately 0.3% over most of the witness range. The interval $-2 < \beta_c < -\sqrt{3}$ and its associated wavelength bounds are therefore exact statements of the local $2 - 4 - 6$ effective kernel; general higher constitutive-response terms may shift the corresponding stationary boundaries of the full rational kernel.

The preferred finite-scale minimum remains at positive gradient energy throughout the interior of the corridor; it reaches zero only at the marginal boundary. This is the primary dimensionless scale-selection result. It selects a preferred structured-response scale without identifying that scale a priori with the Planck length and without, by itself, proving particle binding. The full corridor and wavelength proof is given in **Supplementary Section D**. Supplementary Code S1 reproduces the local corridor boundaries, preferred dimensionless wave number and wavelength interval directly from the $2 - 4 - 6$ kernel and independently verifies exact-rational-kernel consistency for the explicit microscopic existence class.

Figure 2 shows representative reduced kernels across the no-minimum, stable finite-scale, marginal and unstable regimes, together with the parameter-independent preferred-wavelength interval of the local $2 - 4 - 6$ effective kernel across the open stable corridor.

3.4 Microscopic equilibrium restricts the effective parameter freedom

The finite-scale coefficients are further constrained once the background is required to satisfy the microscopic constitutive equilibrium equations. On the $w = 0$ branch,

$$V'_s(s) + \frac{1}{2}K'_4(s)Q^2 = 0, \quad Q = \Delta\bar{u},$$

so that

$$Q^2 = -\frac{2V'_s(s)}{K'_4(s)}.$$

The local constitutive gap and scalar-constitutive mixing are then

$$D_s = V''_s(s) + \frac{1}{2}K''_4(s)Q^2, \quad \Gamma_s = K'_4(s)Q.$$

Consequently, Q , D_s , Γ_s , $K_{4,eff}$ and $K_{6,eff}$ are correlated outputs of the same on-shell branch rather than independently tunable microscopic constants. The admissible scalar theory therefore occupies a restricted action-derived manifold in effective-coefficient space, and the finite- k corridor selects a subset of that manifold. This removes a substantial part of the phenomenological freedom present when the reduced coefficients are treated independently. The on-shell construction is developed in **Supplementary Section E**.

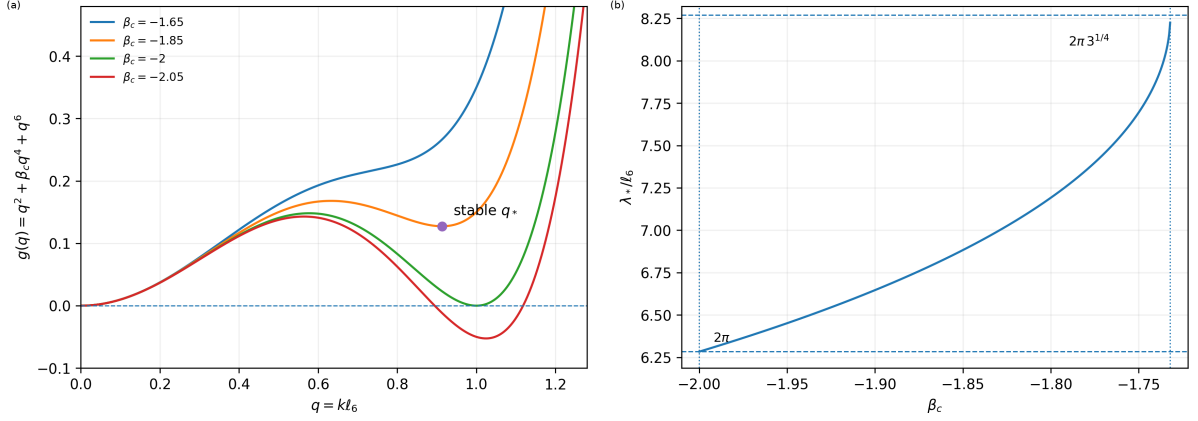


Figure 2: Stable finite-scale corridor generated by the reduced 2 – 4 – 6 kernel. Panel (a) shows representative dimensionless gradient kernels. For $\beta_c = -1.65$ no distinct finite- q stationary pair is present; $\beta_c = -1.85$ lies inside the stable finite-scale corridor and develops a positive non-zero minimum q_* ; $\beta_c = -2$ is the marginal boundary at which the finite- q minimum reaches zero; and $\beta_c = -2.05$ lies beyond marginality and develops a negative minimum. Panel (b) shows the preferred wavelength throughout the open stable corridor. Within the local 2 – 4 – 6 effective kernel, the bounds are parameter-independent once lengths are measured in units of the sixth-order crossover scale. The corresponding minimum is independently confirmed in the unexpanded rational Schur kernel for the explicit microscopic existence class; see Supplementary Section C. Curves are generated directly by Supplementary Code S1.

3.5 Primitive C_3 symmetry and its lowest anisotropy yield exactly three internal orientations

For the bounded internal potential

$$V_c(s, w) = V_s(s) + \frac{1}{2} M_w^2(s) |w|^2 - \kappa_3 \text{Re}(w^3) + \frac{\lambda_w}{4} |w|^4,$$

write $w = r e^{i\theta}$. The angular dependence becomes $-\kappa_3 r^3 \cos(3\theta)$. For $\kappa_3 > 0$ the stable orientations are

$$\theta_n = \frac{2\pi n}{3}, \quad n = 0, 1, 2.$$

Changing the sign of κ_3 shifts the three orientations collectively by $\pi/3$ but does not alter their multiplicity. The finite-amplitude phase has positive radial and angular gaps and is separated from the symmetric phase by a first-order transition. Thus the count of three is not inserted through three primitive family fields: it follows from the primitive C_3 constitutive symmetry and its lowest anisotropic invariant. The result supplies a three-state family-like multiplicity, but it does not determine the observed mass hierarchy, mixing matrices or CP phases. The complete phase analysis is given in **Supplementary Section F**.

3.6 The RSTM scalar equation re-emerges as a collective law

After the constitutive variables are relaxed, the leading local scalar equation takes the effective form

$$\rho_{eff} \ddot{u} - T_{eff} \Delta u + K_{4,eff} \Delta^2 u - K_{6,eff} \Delta^3 u + V_{eff,u} = 0.$$

For a homogeneous linear background with $m_{eff}^2 = V_{eff}''$, the corresponding dispersion relation is

$$\omega^2(k) = \frac{1}{\rho_{eff}} \left[m_{eff}^2 + T_{eff} k^2 + K_{4,eff} k^4 + K_{6,eff} k^6 \right].$$

The familiar second-, fourth- and sixth-spatial-derivative RSTM hierarchy therefore survives, but its status changes: it is a collective equation obtained after microscopic constitutive reduction. Within the stable finite-scale corridor $K_{4,eff}$ may be negative while the full mode energy remains positive. Former universal damping and external forcing terms are not part of this closed equation; such terms belong only to explicitly reduced, driven or open-system descriptions. The action-to-effective-scalar descent is given in **Supplementary Sections A-C**, with the finite-scale implications in **Supplementary Sections D-E**.

3.7 Exact fixed-Noether-charge matter localisation and stable face branch

3.7.1 A globally positive face parent generates the required lower-face response

The matter sector is supplied by the conservative two-face occupation action rather than by interpreting the finite- k scalar minimum as a particle. Its mass operator belongs to the positive-square class

$$M_F^2 = m_f^2 I + P^2,$$

with Hermitian P . Hence

$$M_F^2 \geq m_f^2 I$$

globally. The attractive even lower-face response required for binding, the odd inter-face transition and the positive even completion are generated by the same parent square. They are therefore not independent corrective terms added after the fact. The global face-parent construction is given in **Supplementary Section A**, with the coercivity audit in **Supplementary Section B**.

3.7.2 Fixed charge produces a one-orbital localised branch

The exact common face-phase symmetry gives the conserved Noether charge Q_F . At fixed charge, the occupied lower face and the common constitutive coordinate satisfy reciprocal static equations, and localisation is judged against the dispersed continuum carrying the same Q_F . For the representative conservative existence-class control, the localised branch becomes energetically preferred at

$$Q_F^{coex} \simeq 79.0,$$

while the first additional $l = 1$ spatial orbital appears at approximately

$$Q_F^{\uparrow=1} \simeq 125.2.$$

The resulting benchmark one-spatial-orbital window is therefore

$$79.0 \lesssim Q_F \lesssim 125.2.$$

At the interior benchmark $Q_F = 100$,

$$\omega \simeq 0.933,$$

and the normalised energy per charge relative to the same-charge continuum is

$$\frac{E}{Q_F} \simeq 0.989 < 1.$$

Along the branch, numerical continuation satisfies the fixed-charge variational identity

$$\frac{dE}{dQ_F} = \omega$$

within numerical accuracy. These values establish a finite benchmark existence window; Q_F in this control calculation is not the dimensional elementary action J_* . The fixed-charge branch and continuum comparison are given in **Supplementary Section G**. The quoted coexistence and first-additional-orbital thresholds are obtained by linear interpolation between adjacent points of the $\Delta Q_F = 2$ continuation scan; they are not separate nonlinear root solves.

3.7.3 Radial, non-radial and full-face stability controls

The exact two-quadrature circular branch remains stationary under direct conservative radial evolution and survives a finite radial perturbation. Linearisation in angular sectors finds no exponentially growing lower/common mode through

$$l = 12$$

at the $Q_F = 100$ interior benchmark, and no such mode through $l = 6$ across the tested one-orbital charge interval. The phase Goldstone mode and the translational Goldstone sector are recovered [42]; the small translation eigenfrequency moves towards zero with increasing resolution while its eigenvector approaches the translation template, providing an independent numerical consistency check.

The dormant complementary-face/scalar block also admits an open Schur-coercive stability corridor. At the representative $Q_F = 100$ benchmark, a vacuum odd response $g_{odd}^{(0)} = 0.5$ gives

$$\lambda_{Schur,min} \simeq 1.78 > 0,$$

while the corresponding critical benchmark odd response is approximately

$$g_{odd,crit}^{(0)} \simeq 1.35.$$

These values are stability diagnostics rather than physical parameter predictions. Taken together, the results establish a benchmark three-dimensional localised matter carrier that is below the same-charge continuum and stable in the tested radial, non-radial and full-face sectors. The physical coefficients must still be derived from the microscopic action and tested blindly against this open stability corridor. The complete stability analysis is given in **Supplementary Section G**. Supplementary Code S2 reproduces the fixed-charge continuation and the radial, non-radial, translation-mode, charge-window and full-face stability diagnostics.

The principal fixed-charge existence and stability results are summarised in Figure 3, including the representative localised profile, the one-orbital charge window, the fixed-charge variational residual and the tested non-radial spectrum.

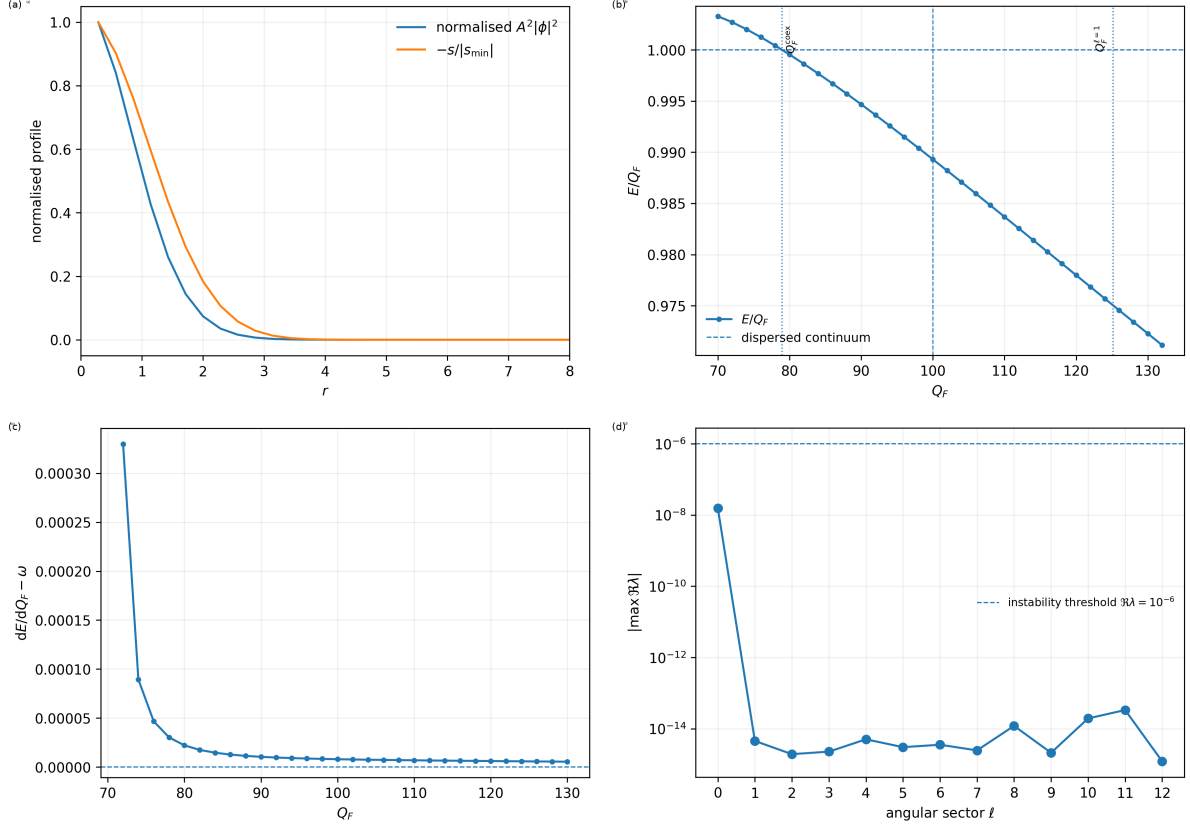


Figure 3: Fixed-Noether-charge one-orbital matter branch and stability controls. Panel (a) shows the $Q_F = 100$ radial structure from the final $N = 90$ positive-square continuation, with the occupation density and common constitutive response separately normalised to display their co-localisation; this final square-parent run gives $\omega = 0.933293$. Panel (b) shows the $N = 100$ fixed-charge continuation: the localised branch crosses below the dispersed same-charge continuum at $Q_F \simeq 79.0$, while the first additional spatial orbital enters at $Q_F \simeq 125.2$, defining the benchmark one-orbital interval. At $Q_F = 100$ the continuation gives $E/Q_F = 0.989308$ and $\omega = 0.934014$. The quoted thresholds are linear interpolations between adjacent $\Delta Q_F = 2$ continuation points. Panel (c) verifies the fixed-charge variational identity through the small continuation residual $dE/dQ_F - \omega$. Panel (d) shows the maximum real part of the co-rotating lower/common stability spectrum through angular sectors $\ell = 0, \dots, 12$; no mode exceeds the adopted instability criterion $\text{Re } \lambda > 10^{-6}$. The small difference 0.934014 to 0.933293 is the documented grid/continuation-resolution shift on the same lower-face mass law, not a different particle branch. Numerical panels are reproduced by Supplementary Code S2.

3.7.4 Covariant phase-frame attachment supplies the reduced spin structure

The localised occupation field and the phase frame are distinct. With

$$\Upsilon = U_F \psi, \quad U_F \in SU(2),$$

and moving-frame connection

$$\Gamma_\mu = -(\partial_\mu U_F) U_F^\dagger,$$

pure orientation is covariantly isospectral to the texture-free occupation operator. The phase frame therefore does not create the binding well. Instead, the fixed- Q_F amplitude/mass sector supplies localisation, while the normalised degree-one frame carries the double-cover topology.

Spinning charge-supported solitons and topological routes to fermionic rotational character are independently established [13,15]. The result here is narrower: the normalised RSTM frame can carry the spin-cover topology while remaining covariantly isospectral to the independently localised fixed- Q_F occupation branch, so spin topology neither supplies nor repairs the particle-binding mechanism.

Transport of the physical rank-two matter fibre gives

$$W(2\pi) = -I_2,$$

and the associated isolated one-particle pole has positive residue,

$$Z_{core} = 1 > 0.$$

A separate positive frame functional possesses a finite Derrick-stable scale [43], and constitutive dependence of the frame stiffness can provide positive centre locking to the matter core. The microscopic descent of the normalised frame and a fully action-derived combined amplitude-frame solution remain open; the existence and tested stability of the fixed-charge occupation branch itself do not. The phase-frame and spin-fibre construction is given in **Supplementary Section H**.

3.8 Relational family response admits a stable fixed-charge flavour-sensitive branch

The threefold multiplicity of Section 3.5 does not itself produce family hierarchy. The additional question is whether a relational fold response can differentiate the internal states while remaining dynamically compatible with the already-localised fixed- Q_F matter carrier.

At the present level, the non-zero fold-face response required to transmit this family information is symmetry-allowed but is not forced by fold symmetry and is not generated by the presently specified reflection-even mediators. Its physical value must therefore descend from deeper fold/interface dynamics or be determined independently of flavour data.

Starting from the symmetric fixed-charge branch, the coupled $\tau_2/u/\delta$ sector develops a finite radial threshold. In the representative normalised control with $k_\delta = 0.1$, the full radial calculation gives

$$g_{odd}^{crit} = 1.1600959962$$

below the corresponding u-only critical value. The critical eigenvector contains finite face, u and family-relative components, showing that the instability is one coupled odd mode rather than three separately imposed mechanisms.

The fixed-charge Lyapunov-Schmidt reduction [44] gives

$$\beta_{eff} = 1.952553647 \times 10^{-3} > 0$$

so the representative bifurcation is supercritical. An independent constrained relaxed-energy reconstruction reproduces this coefficient with

$$\frac{|\beta_{eff}^{fit} - \beta_{eff}^{LS}|}{|\beta_{eff}^{LS}|} = 4.39 \times 10^{-7}$$

providing a separate numerical cross-check.

Direct nonlinear continuation then releases the even radial profiles and follows the non-zero branch. The full fixed- Q_F calculation continues successfully over

$$1.0005 \leq g/g_c \leq 1.20$$

with lower constrained energy than the symmetric branch throughout the tested interval and positive radial curvature. The family-relative component remains finite after the occupied particle carrier is allowed to relax.

The complete co-rotating dynamical problem retains both face quadratures together with the u , family-angle and even constitutive responses. No non-symmetry exponentially growing mode is found through the tested sectors

$$\uparrow=0, \dots, 12$$

The common-phase Goldstone direction is recovered, while the small $\ell = 1$ mode moves towards zero with grid refinement and its eigenvector approaches the translation template. The latter behaviour identifies the residual frequency as finite-grid translation-symmetry splitting rather than a physical instability.

These calculations establish an action-class numerical-control result. They show that the minimal relational architecture possesses a non-empty stable nonlinear corridor capable of carrying family-sensitive response without destroying the fixed-charge particle. The numerical values of c_Y , ϵ_{sw} , Π_Y and other benchmark coefficients are not identified with their physical values.

Once the corresponding physical responses are independently descended, the three family-conditioned matter profiles define sector-specific operators M_f . Their Hermitian products

$$H_f = M_f M_f^\dagger$$

determine the left-handed diagonalising bases, from which

$$V_{CKM} = U_u^\dagger U_d, \quad U_{PMNS} = U_e^\dagger U_\nu$$

follow in the conventional manner. These relations define the downstream prediction step and are not used in the benchmark calculation.

Figure 4. Relational family response and benchmark fixed-charge flavour-sensitive branch

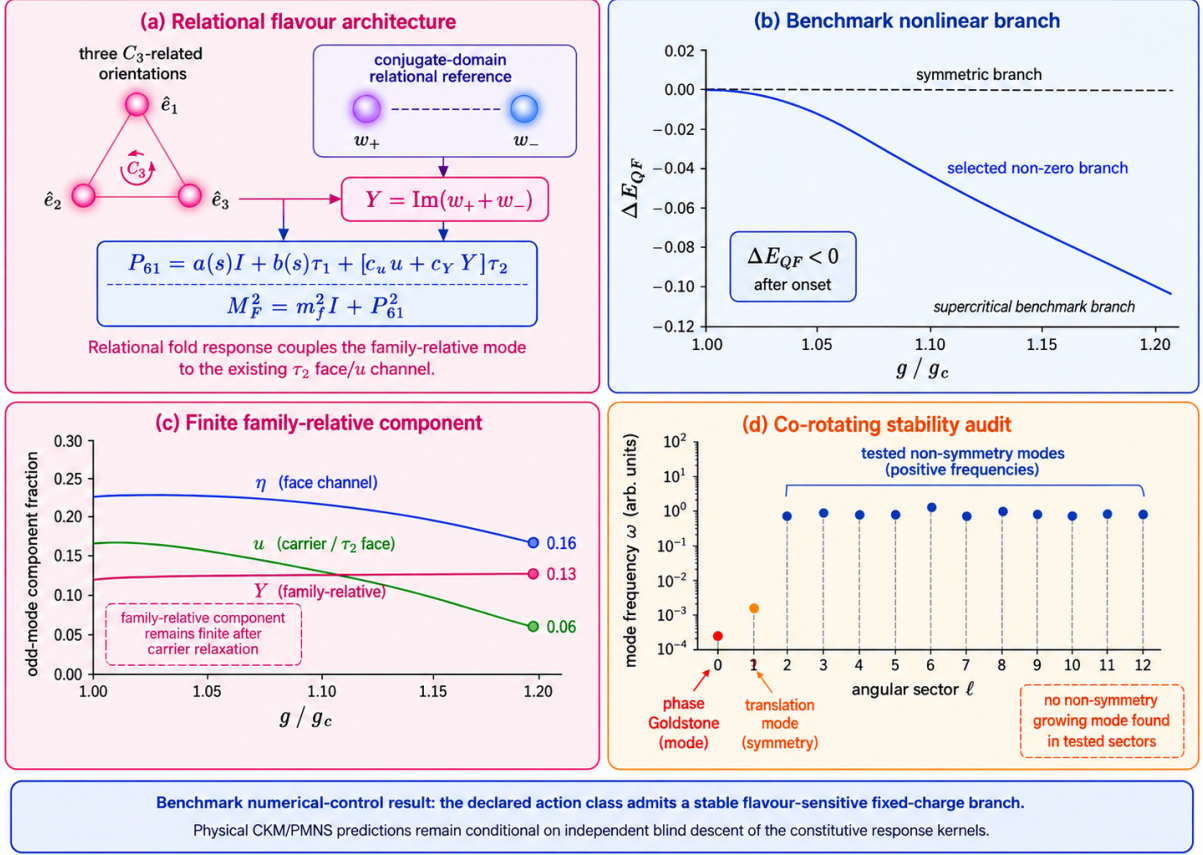


Figure 4: **Figure 4. Relational family response and benchmark fixed-charge flavour-sensitive branch.** Panel (a) summarises the action-class relational architecture: three C_3 -related orientations are distinguished relationally by the independently responding conjugate-domain quantity $Y = \text{Im}(w_+ w_-)$, where w_+ and w_- denote the plus/minus conjugate-domain responses. This quantity enters the same positive-square τ_2 face axis as the microscopic u coordinate. Panel (b) shows the full $N = 90$ fixed- Q_F continuation; the selected branch has lower constrained energy than the symmetric branch at every tested point. Panel (c) shows the odd-mode component fractions along the same continuation; the family-relative component remains finite after carrier relaxation. Panel (d) shows the full co-rotating non-symmetry stability audit at $N = 44$ and $g/g_c = 1.20$: no mode crosses the adopted instability criterion through $\ell = 12$. These are benchmark/action-class controls, not physical flavour-parameter predictions. Physical CKM/PMNS predictions remain conditional on independent blind descent of the response kernels. The numerical panels and stability controls are reproduced by Supplementary Code S5.

3.9 Positive-frequency reduction generates Schrödinger-like dynamics

The real conservative face system possesses positive- and negative-frequency branches. Restricting to the stable positive-frequency sector equips the canonical real phase space with its natural complex structure and gives

$$iJ_* \partial_t \psi = H_{eff} \psi.$$

For the exact circular face carrier, the phase, spectral and symplectic definitions of the classical carrier action coincide:

$$J_{phase} = J_{spectral} = J_{symplectic} = Q_F.$$

Thus the action carried by the circular matter branch is no longer an unspecified analogy: it is the exact Noether action of the common face phase. What remains open is the absolute universal elementary normalisation J_* . A global rescaling of the classical action leaves the Euler-Lagrange trajectories unchanged while rescaling momenta, energies and action variables, so $J_* = \hbar$ cannot be obtained by choosing the minimum of an arbitrary classical orbit. The positive-frequency construction and action identity are developed in **Supplementary Section H**.

3.10 Conditional fermionic mode algebra follows from the odd exchange sector

The independent two-particle exchange calculation gives an odd relative exchange character in the separated-constituent limit. This result is distinct from the one-particle 2π rotation sign. Conditional on the complete microscopic dynamics selecting that odd one-particle sector, the multiparticle space is the antisymmetric exterior Fock space,

$$\mathcal{F}_-(\mathcal{H}_1) = \bigoplus_{N=0}^{\infty} \wedge^N \mathcal{H}_1.$$

For orthonormal one-particle modes, the creation and annihilation operators then obey

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \quad \{a_i, a_j\} = 0, \quad \{a_i^\dagger, a_j^\dagger\} = 0.$$

The mode CAR are therefore an algebraic consequence of antisymmetric second quantisation once the odd exchange representation is physically selected; they are not an additional microscopic Grassmann postulate. The primitive RSTM variables remain commuting fields. Compulsory microscopic odd-sector selection and the subsequent local $Spin(1, 3)$ -covariant, particle/antiparticle-complete and graded-microcausal field algebra remain open. The exchange and CAR derivation is given in **Supplementary Section I**.

3.11 Internal-frame covariance constrains the global charge classes

Local covariance of the internal oscillator frame supplies a structural gauge-connection architecture. Conditional on faithful action of that oscillator/internal frame, the globally acting bundle is constrained to

$$G_{faith}^{RSTM} = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\mathbb{Z}_6},$$

with charge-class congruence

$$Y \equiv \frac{\sigma}{2} - \frac{\tau}{3} \pmod{1}.$$

The $\mathbb{Z}_6$ quotient has a recognised Standard Model precedent [39]. This is stronger than merely observing that gauge-like matrices can be written down: the faithful global action restricts the allowed bundle and charge classes. It remains a structural/conditional result rather than a completed microscopic Yang-Mills derivation [40]. Local physical gauge redundancy, the gauge kinetic normalisations and the physical couplings g_1 , g_2 and g_3 remain to be obtained from the action. The continuity and claim audit for this result is given in **Supplementary Section O**, Section O.5.

3.12 Chirality arises separately from the two-face/orientation sector

The chiral structure is not identified with either the spin-cover sign or the exchange sign. At the effective level the two-face/orientation sector supports the projector architecture

$$P_W = F_+ P_L + F_- P_R.$$

This gives a visible-face construction compatible with left-handed weak currents and a chirality-reversed complementary branch. Face reflection remains an internal operation and is not fermion parity. The result is therefore a chiral precursor compatible with the required handedness structure, not a completed local chiral Standard Model. Its combination with the reduced spin fibre and the emergent spacetime tetrad into a local relativistic matter field remains open. The chiral construction is given in **Supplementary Section H**.

3.13 Born-form channel weights and representative deterministic detector recording

Linear projection of a positive-frequency state onto detector channels gives amplitudes

$$A_i = \langle d_i | \psi \rangle.$$

Quadratic detector energy response then gives the normalised channel weights

$$P_i^{(wt)} = \frac{|\langle d_i | \psi \rangle|^2}{\sum_j |\langle d_j | \psi \rangle|^2},$$

or, for a general positive detector operator E_i ,

$$P_i^{(wt)} = \langle \psi | E_i | \psi \rangle.$$

A deterministic threshold detector was then used to test whether unresolved microscopic detector states could sample these pre-existing quadratic intensities. Fitting the observed outcome frequencies to a power-law sampling exponent gave

$$p = 1.038,$$

with bootstrap interval

$$0.967 < p < 1.181,$$

and

$$RMSE = 0.0517.$$

A negative control in which the required intensity dependence was removed drove the fitted exponent to the lower search bound

$$p = 0.050.$$

The representative detector family therefore sampled the quadratic channel weights approximately linearly, whereas the intensity-independent control did not. The detector-control parameters were selected by an explicit finite scan targeting $p \simeq 1$; the result is therefore a calibrated representative existence benchmark rather than an independent prediction. The Lorenz/SRB microdynamics used to sample unresolved detector states is a reduced deterministic dissipative detector proxy, not an identification of the fundamental conservative RSTM substrate dynamics. Damping is not used to create the quadratic analyser weights: it enters only in the reduced detector/environment description to provide absorption, amplification, decoherence and persistent recording while the complete microscopic system remains conservative.

This establishes Born-form channel response and a representative deterministic recording mechanism. It does not yet establish the universal physical preparation measure over unresolved detector and state variables. For detector basins $B_i(\psi)$, the remaining condition is

$$\mu_{prep}[B_i(\psi)] = P_i^{(wt)}$$

for arbitrary physical preparations and measurement contexts. The full detector-basin analysis is given in **Supplementary Section J**. Supplementary Code S3 supplies the complete detector implementation, parameter scan, bootstrap analysis, negative control and reference outputs. The unresolved-state ensemble need not erase the coherence of the prepared collective matter state. A fixed rank-two spin preparation may retain its relative phase and spin-frame orientation while differing in microscopic variables not specified by the preparation. The detector calculation should therefore be interpreted as sampling unresolved microstates conditional on a coherent prepared state, rather than as randomising the state whose analyser amplitudes generate the quadratic channel weights. The invariant compact-phase measure identified in Supplementary Section J provides a symmetry-selected candidate component of that unresolved ensemble; universal physical sampling of the complete preparation measure remains open. The positive-frequency reduction, the separation between quadratic analyser weights and detector sampling, the representative deterministic recording benchmark and its negative control are summarised in Figure 5.

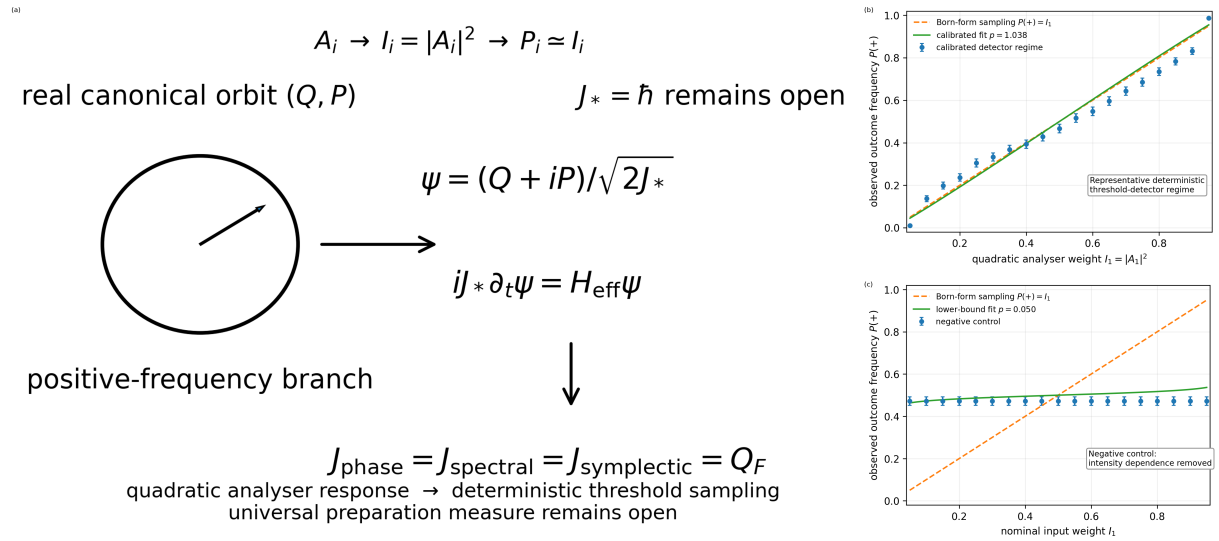


Figure 5: Positive-frequency complex reduction and representative deterministic detector benchmark. Panel (a) illustrates the canonical reduction of a stable real oscillatory mode to its positive-frequency complex description. On the exact circular face carrier, the phase-generator, spectral and symplectic action definitions coincide with Q_F ; the absolute universal elementary action J_* remains to be determined and is not imposed equal to $\hbar$. The upper chain separates the two stages relevant to Born-form measurement: analyser amplitudes A_i first give quadratic analyser weights $I_i = |A_i|^2$, after which a detector must sample those pre-existing weights with outcome frequencies P_i . Panel (b) shows the recovered deterministic SRB/Lorenz threshold-detector benchmark against the quadratic analyser weight $I_1 = |A_1|^2$. The Born-form comparison is therefore the linear sampling relation $P(+) = I_1$, not a second squaring of I_1 . A finite control-parameter scan targeted $p \simeq 1$ and selected a representative regime giving $p = 1.038$, 95% bootstrap interval 0.967–1.181 and RMSE = 0.0517. The result is therefore a calibrated representative existence benchmark, not an independent prediction or universal Born theorem. Panel (c) removes the required intensity dependence; the fitted exponent then reaches the lower search bound $p = 0.050$ and the Born-form sampling relation is lost. The Lorenz/RK4/SRB dynamics is a reduced deterministic detector proxy rather than the fundamental closed conservative RSTM microdynamics. The universal preparation measure required to make $P_i \simeq I_i$ context-independent remains open. Numerical detector panels and the negative control are reproduced by Supplementary Code S3.

3.14 Incompatible collective strain reproduces spatial Einstein curvature

The primitive scalar u is not identified directly with the gravitational metric. Instead, scalar and symmetric-traceless quadrupole moments of the microscopic state form a locally complete collective spatial metric. At linear order,

$$h_{ij} = 2Q_{ij} + 2\phi\delta_{ij},$$

which contains the six independent components of a symmetric spatial metric.

Ordinary compatible strain generated by a displacement has vanishing Saint-Venant incompatibility and is therefore interpreted as a change of ruler/reference description. For an independent incompatible collective strain ε_{ij} , with $h_{ij} = 2\varepsilon_{ij}$, the result is

$$G_{ij}^{(1)}[h = 2\varepsilon] = Inc(\varepsilon)_{ij}.$$

The linearised spatial Einstein tensor therefore appears directly as the incompatibility of the collective strain [18]. This sharpens the earlier strain-metric intuition: gravitational curvature is not ordinary material displacement of the membrane but an incompatible collective deformation of its microscopic state. The construction is derived in **Supplementary Section K**.

3.15 Spatial reference redundancy fixes the Einstein coefficient ratio

The collective metric decomposition admits the spatial relabelling

$$\delta h_{ij} = 2\partial_{(i}\xi_{j)},$$

under which compatible reference deformations lie in null directions of the physical curvature kernel. Requiring the most general parity-even quadratic two-derivative spatial action to respect this Ward identity fixes the four apparent coefficients to the ratio

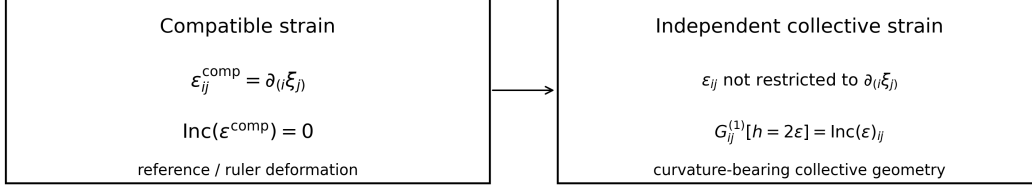
$$a : b : c : d = 1 : -2 : 2 : -2.$$

Thus the spatial Einstein operator is selected by collective reference redundancy rather than by tuning four independent coefficients. Only one overall normalisation remains at this level.

This is a structural spatial gravity result. The constructive scalar-vector-quadrupole hierarchy can assemble the required operator through elimination of a healthy auxiliary collective sector, but descent of the target coefficients from the actual microscopic action is not yet complete. The physical common transverse-traceless pole, its positive residue and propagation cone, the metric-normalised gravitational coupling, temporal Fierz-Pauli recombination and nonlinear hypersurface-deformation algebra remain open. The Ward derivation and these boundaries are given in **Supplementary Section K**. Supplementary Code S4 independently verifies the linear Einstein/incompatibility identity and the Ward-null coefficient ratio symbolically.

Figure 6 summarises the separation between compatible reference deformation and curvature-bearing incompatible collective strain, together with the Ward selection of the spatial Einstein coefficient ratio.

(a)



The primitive scalar u is not itself the metric scalar.
Geometry is a collective incompatible-strain construction.

(b)

Most general parity-even two-derivative spatial kernel

$$\mathcal{L}_\nabla = a(\partial Q)^2 + b(\partial \cdot Q)^2 + c(\partial \cdot Q) \cdot \partial \phi + d(\partial \phi)^2$$

↓

spatial relabelling / compatible directions are Ward-null

Figure 6: Collective incompatible strain and Ward-selected spatial Einstein structure. Panel (a) separates ordinary compatible strain, which is generated by a displacement field and has vanishing Saint-Venant incompatibility, from the independent collective strain used to carry spatial curvature. At linear order the incompatibility of the collective strain is the linearised spatial Einstein tensor for $h = 2\epsilon$. Panel (b) starts from the most general displayed parity-even quadratic two-derivative spatial kernel. Requiring compatible spatial relabellings to be Ward-null fixes the relative coefficients uniquely up to one overall normalisation. This is a structural spatial result: it does not by itself establish the physical tensor pole, temporal completion, nonlinear constraint algebra or full four-dimensional GR. The algebraic identity and coefficient ratio are independently checked by Supplementary Code S4.

$$a : b : c : d = 1 : -2 : 2 : -2$$

3.16 The selected finite microscopic truncation has controlled ultraviolet behaviour

At sufficiently large wavenumber the scalar dispersion is dominated by the positive sixth-order term, [21]

$$\omega^2 \simeq \frac{K_{6u}}{\rho_u} k^6, \quad \omega \propto k^3,$$

corresponding to Lifshitz-type anisotropic scaling [21] with

$$z = 3.$$

For the declared finite u, s, w truncation, the sixth-order principal operator is marginal while the displayed lower-derivative and local interaction vertices carry positive weighted coupling dimension. The selected finite model is therefore weighted super-renormalisable at this stated scope [22]. The improvement is achieved with higher spatial derivatives while the fundamental microscopic time evolution remains second order [23].

For the stable free scalar Euclidean covariance, reflection of the preferred Euclidean time gives a positive quadratic form and therefore a positive anisotropic Hamiltonian reconstruction. Importantly, a negative intermediate $K_{4,eff}$ inside the stable finite-scale corridor does not violate this result; the relevant condition is positivity of the complete mode energy.

The scope is deliberately limited. Because the scalar and constitutive fields have zero weighted dimension, arbitrary powers remain allowed by weighted counting; $z = 3$ does not by itself select a unique finite exact functional theory. Nor does preferred-time reflection positivity establish full Euclidean rotational invariance or a Lorentz-invariant Osterwalder-Schrader reconstruction [24,25]. Complete coupled RSTM renormalisability, the fundamental fermion/gauge ultraviolet operators, radiative protection of the infrared Lorentz surface and the ultimate ultraviolet causal completion remain open. The complete audit is given in **Supplementary Section L**.

3.17 Positive sixth-order energy regulates zero-width cores and retains an effective area law

For a localised configuration with characteristic amplitude U and radius R , the sixth-order energy scales as

$$E_6 \sim \frac{K_6 U^2}{R^3}.$$

For $K_6 > 0$,

$$E_6 \rightarrow \infty \quad \text{as} \quad R \rightarrow 0$$

at fixed non-zero amplitude. The microscopic theory therefore possesses a genuine energetic obstruction to unrestricted zero-width scalar collapse. A finite equilibrium core still requires competition with the remaining binding, source and nonlinear terms; the sixth-order term alone does not determine its radius.

The retained effective/core entropy calculation gives

$$S_{core} \simeq \frac{k_B A_*}{4} \left[1 + O\left(\frac{\lambda_c}{R_*}\right) \right],$$

and hence

$$S_{core} \propto A_*$$

for $\lambda_c \ll R_*$. This is an achieved area-law result at the stated effective/core level. It is not an independent microscopic count of all physical horizon states, and its coefficient must not be fed back as an assumption into such a future count. Likewise, finite microscopic/core regulation is not by itself a theorem of metric regularity, bounded spacetime curvature or geodesic completeness. The separate Wald/Bekenstein route remains conditional on independently completed gravitational and action normalisations. These distinctions are developed in **Supplementary Section M**.

3.18 A normalisation-independent relation links the length, action and gravity sectors

The microscopic crossover/ruler scale $\uparrow_*$, elementary action scale J_* and collective gravitational coupling G_{RSTM} are defined in different sectors and are kept independent. Their physical values must be obtained from their respective sector calculations rather than inferred from one another. Their natural dimensionless combination is

$$C_* = \frac{c^{3\uparrow_*^2}}{J_* G_{RSTM}}.$$

Using the metric-normalised tensor stiffness Z_ϵ and the dimensionless definitions ζ_ϵ and $\hat{J}_*$ gives the equivalent closure

$$C_* = \frac{8\pi\zeta_\varepsilon}{\hat{J}_*}.$$

A global multiplication of the classical microscopic action leaves the Euler-Lagrange trajectories unchanged, while J_* and the inverse gravitational coupling scale in compensating ways. The combination C_* is therefore invariant under this otherwise arbitrary overall action normalisation. It probes cross-sector physics rather than a convention for units or classical action scale.

This relation does not determine any of its three physical inputs from the other two. Instead, $\uparrow_*$, J_* and G_{RSTM} must each be obtained from its own microscopic calculation and then compared blindly with the conventional Planck length, $\hbar$ and Newton's constant. The Planck relation is therefore a possible downstream closure result rather than a microscopic definition. The invariant and its action-rescaling audit are derived in **Supplementary Section N**.

The six headline results are therefore Structure, Scale, Particle, Family, Quantum and Geometry; the normalisation-independent cross-sector invariant provides the closing synthesis.

4. Discussion

The results above change the status of RSTM more substantially than a new parameter choice or an additional phenomenological sector would. The earlier formulation asked how many familiar physical structures could be represented within one higher-gradient membrane equation. The present construction asks the more restrictive question of which of those structures can be obtained within an action-first hierarchy after admissible states have been selected and unresolved information has been reduced. This section discusses that common mechanism, the continuity with the earlier RSTM formulation, and the principal tests that now separate structural success from a completed physical theory.

4.1 Six headline results and a common emergence mechanism

The six headline results — Structure, Scale, Particle, Family, Quantum and Geometry — are not six independent constructions appended to one microscopic model. Their significance lies partly in the repeated route by which they arise. Across the sectors developed here, the framework uses a common action-first ordering: the microscopic action first supplies the explicitly admitted dynamical sectors and their allowed configurations. Stability, equilibrium, conserved charge or topology then selects the relevant collective branch. Only after that selection are constitutive, environmental or reference degrees of freedom relaxed, projected or quotiented.

$$S_{micro} \rightarrow \text{dynamical selection} \rightarrow \text{conservative reduction} \rightarrow \Gamma_{eff}.$$

$$\Gamma_{eff} \rightarrow \text{effective symmetry/law} \rightarrow \text{observable}.$$

This common action-first ordering is the principal unifying feature of the microscopic reconstruction.

In the scalar sector, a healthy common constitutive coordinate is eliminated only after the curvature-populated branch is specified; the Schur complement then lowers the relaxed fourth-order response while increasing the sixth-order regulator. In the matter sector, the positive face parent supplies an exact phase symmetry and conserved Noether charge before fixed-charge reciprocal binding selects the one-orbital branch. In the family sector, the three C_3 orientations are first formed by the constitutive anisotropy and only then distinguished relationally through the conjugate-domain response and its coupled odd branch. The phase frame is attached only after localisation, so topology does not have to perform the work of binding.

Each of the broad ingredients in this matter construction has precedent when considered separately or in other combinations. Q-balls establish localisation at fixed Noether charge [11]; Q-lumps

demonstrate coexistence of conserved charge and topology [12]; spinning Q-balls demonstrate rotating charge-supported solitons in $3 + 1$ dimensions [13]; higher-spatial-derivative theories can support topological Q-ball solutions [14]; and Finkelstein-Rubinstein-type constructions demonstrate topological routes to half-integer rotational and exchange behaviour [15]. The RSTM claim is therefore not priority for these ingredients. Its distinctive result is their dynamically ordered separation: constitutive Schur relaxation first selects a finite structural scale but is shown insufficient for binding; a separate fixed- Q_F two-face sector then produces the localised carrier; and a distinct normalised frame subsequently supplies spin-cover topology without altering the binding spectrum. Exchange is calculated independently again rather than inferred from the one-particle rotation sign.

The same ordering recurs downstream. Positive-frequency selection converts a real conservative canonical system into a complex quantum-like description; detector reduction turns deterministic channel dynamics into irreversible records; and spatial reference directions are quotiented from the collective strain sector before the Einstein curvature operator appears. The repeated mathematical operation is therefore not the insertion of a new fundamental law at each level but the controlled loss of microscopic information after dynamical selection. This is the sense in which the present paper uses the term emergence by conservative reduction. The consolidated hierarchy and status language are given in Supplementary Section P.

4.2 Disposition of the earlier RSTM claims

The earlier RSTM formulation is neither retained wholesale nor treated as a discarded model. Its role was to make a broad set of proposed connections explicit enough to test. The present microscopic reconstruction retains the structures that survive those tests, reinterprets those that belong at an effective level, and removes or narrows claims whose earlier interpretation was stronger than the available derivation. Supplementary Section O gives the detailed claim-by-claim continuity audit.

The second-, fourth- and sixth-spatial-derivative scalar hierarchy is retained, but as a constitutively reduced collective equation. The earlier single-carrier Schrödinger-like envelope construction is retained and sharpened into an explicit positive-frequency reduction one level further downstream. The earlier Planck-anchored parameter practice is superseded by independent microscopic derivation and blind comparison. The bimodal spinor intuition is strengthened into a canonical positive-frequency rank-two fibre with independently calculated rotational and exchange structure, while compulsory odd-sector selection and the local relativistic field remain open.

Gauge-like rephasings survive as internal-frame covariance and a constrained global bundle, but physical Yang-Mills dynamics and couplings have not yet descended from the action. The representative Born-compatible detector result survives, while the universal preparation measure is isolated as the remaining statistical problem. The earlier strain-metric intuition is retained in a sharper form: compatible strain is reference-like, incompatible collective strain carries spatial Einstein curvature, and a Ward identity fixes the two-derivative coefficient ratio. Direct identification of scalar stiffness with Newton's constant is not retained.

Higher-gradient ultraviolet regulation, finite-core resistance to collapse and the effective/core area-law result also survive at their stated levels. They no longer imply, by themselves, complete singularity resolution, Hawking radiation, greybody factors or microscopic horizon-state counting. Likewise, the three-family and flavour calculations remain useful, but the old three-primitive-field ontology is replaced by the minimal C_3 constitutive sector, and the fitted CKM/PMNS scans remain phenomenological consistency demonstrations until the physical family operators are derived. The ultraviolet result is now explicitly restricted to the selected finite truncation rather than claimed as a completed renormalisability and all-scale Osterwalder-Schrader theorem. These changes are summarised in Supplementary Section O.

4.3 From mirror domain to CPT-folded conjugate sector

The conjugate-domain architecture remains a central element of RSTM, but its role is now more precisely separated from the effective structures built on it. The two faces are treated as complementary branches of one conservative microscopic completion with opposite temporal orientation across the fold. A contribution that appears negative in a one-face bookkeeping convention can therefore represent redistribution between sectors of the closed system rather than a fundamentally unbounded negative Hamiltonian.

This architecture provides a natural setting for matter/antimatter conjugacy and complementary chirality, but these interpretations are not yet equivalent to a completed local relativistic antiparticle construction. Face parity is not fermion parity and does not determine the exchange sign. The visible and complementary branches may also support anomaly-consistent or vector-like high-energy completions, but anomaly consistency does not itself derive the physical spectrum. The corresponding local and global Standard Model anomaly constraints are established independently [45,46].

The same caution applies to vacuum physics. The present theory distinguishes microscopic fluctuation energy, constitutive stiffness renormalisation, local matter-conditioned residual energy and a homogeneous vacuum residual. They may ultimately share a microscopic origin, but they are not interchangeable variables. In particular, the small observed cosmological residual and an equation of state close to -1 remain quantities to be derived rather than recovered by identifying them with the finite-scale stiffness shift. The conjugate-domain continuity and the cosmological claim boundary are given in Supplementary Section O, Section O.10.

4.4 From fundamental damping to detector-local irreversibility

One of the clearest changes from the earlier formulation concerns damping. In v7, a positive Rayleigh term appeared in the master scalar equation and also played a role in decoherence and detector modelling. In the present hierarchy the closed microscopic action is conservative. A universal microscopic damping coefficient is therefore no longer required.

closed conservative dynamics → eliminate hidden/environmental variables

→ memory kernel → local damping approximation.

This does not remove irreversible physics from the effective description. Absorption, decoherence, relaxation and stable detector records can all occur after apparatus or environmental degrees of freedom have been omitted. What changes is the causal ordering. Damping records and stabilises the selected detector branch; it does not create the incoming amplitudes or their squared-overlap weights. For the same reason, phenomenological damping is not used as the microscopic origin of Hawking temperature. The detector reduction is developed in Supplementary Section J and its continuity with the earlier formulation is summarised in Supplementary Section O.

Threshold-detector approaches to Born-compatible statistics have prior precedent [41]; the RSTM distinction is the causal placement of analyser response, deterministic basin competition and effective irreversibility within the conservative-reduction hierarchy.

4.5 The changed status of the scalar PDE and Schrödinger equation

The scalar PDE and the Schrödinger-like equation are both retained, but neither remains at the top of the ontology. The current hierarchy is

$$S_{micro} \rightarrow \Gamma_{eff}[u] \rightarrow 2 - 4 - 6 \text{ scalar PDE}$$

$$\rightarrow \text{positive - frequency sector} \rightarrow iJ_*\partial_t\psi = H_{eff}\psi.$$

This placement resolves two earlier ambiguities. First, the effective scalar coefficients can now be correlated outputs of constitutive relaxation rather than independently calibrated microscopic constants. Secondly, complex quantum evolution does not need to be inserted into the primitive action: it follows after the stable real canonical system has been restricted to a positive-frequency sector.

There is also continuity with the earlier RSTM development. Version 3 had already obtained first-order complex Schrödinger-like envelope dynamics from the real second-order STM equation by selecting a single carrier $e^{i(k_0x - \omega_0t)}$ within a multiple-scale reduction [37]. The present construction makes the frequency-sign structure explicit and replaces that earlier carrier-envelope argument by a canonical positive-frequency reduction on the physical matter branch. Singh subsequently developed an independent and more general positive-frequency formulation in which the temporal Hilbert transform supplies the complex structure on the selected solution subspace [38]. The new RSTM result is therefore neither the generic existence of complexification from real oscillatory dynamics nor a reproduction of Singh's Hilbert-transform construction: it is the action-level placement of the positive-frequency reduction downstream of the independently localised fixed-charge carrier, together with the identification of its phase, spectral and symplectic carrier actions.

The change also sharpens the remaining quantum problem. The fixed face phase identifies an exact carrier action and the phase, spectral and symplectic definitions coincide on the circular branch, but classical orbit geometry does not fix the universal dimensional value of J^* . The comparison $J^* = \hbar$ therefore remains a blind downstream test. The scalar hierarchy is derived across Supplementary Sections A-E, while the positive-frequency and action construction is given in Supplementary Section H.

Rosetta continuity — one microscopic dynamics, three effective dialects

V7 summarised the relation among response, evolution and action descriptions as a “Rosetta — one physics, three dialects” [1]. That organising idea is retained, but its common source is moved one level deeper. In the present hierarchy,

$$S_{micro}[u, s, w, faces, \dots] \rightarrow B_{phys} \rightarrow \Gamma_{eff}$$

where B_{phys} denotes the dynamically selected physical branch. The resulting reduced dynamics may then be represented in response language through $G_R(\omega, k)$, self-energies, poles and residues; in evolution language through the positive-frequency H_{eff} acting on ψ_{pos} ; or in action language through Γ_{eff} and its stationary-phase/path-integral representation. These are related descriptions of the same reduced branch rather than interchangeable microscopic objects. Open detector dynamics constitutes a further reduction and is not part of the closed Rosetta equivalence. The current formulation therefore retains the v7 Rosetta correspondence while replacing the effective scalar PDE as its fundamental source by the conservative microscopic action.

4.6 Spin, chirality, exchange and gauge structure are not one mechanism

Several two-component structures occur in the matter sector, but the present organisation deliberately prevents them from being identified with one another. The constitutive variable w is a family-like redistribution doublet. The positive-frequency matter fibre is a different rank-two object. The two faces provide a separate complementary orientation. The phase frame carries the double-cover rotational topology. Two-particle exchange is a path in the configuration space of separated constituents, and internal-frame covariance is the origin of the structural gauge connection.

This separation is a strength rather than a loss of unification. A 2π rotation sign does not prove fermionic statistics; exchange oddness does not by itself produce a local relativistic spin-statistics theorem [47]; face reflection does not supply fermion parity; and a local connection does not automatically determine the Yang-Mills kinetic action or its couplings. Each link can therefore be

tested independently. Supplementary Sections H and I give the spin and exchange constructions, while Supplementary Section O.5, gives the gauge/chirality continuity audit.

This separation also defines the relevant originality boundary. Existing soliton theories show that conserved charge, topology, finite-size localisation and angular momentum can coexist [12–15]. The narrower RSTM structural result is the separation and ordering of their dynamical responsibilities: finite- k constitutive response selects scale, fixed- Q_F reciprocal dynamics supplies binding, a covariantly isospectral normalised frame supplies the one-particle spin-cover structure, and the two-particle exchange character is evaluated separately on a degree-two configuration-space path.

The remaining odd-sector problem should therefore be understood as a physical-selection problem. The reduced rotational centre character and the independent exchange-odd representation have already been constructed; what remains is to derive why the complete microscopic dynamics selects that available rank-two sector as the elementary physical matter branch.

4.7 Retained family mathematics and replaced ontology

The earlier Y.12 route used a three-basin or triple-well picture to motivate the observed family count. The present microscopic ontology makes three primitive family fields unnecessary. Instead,

$$w \rightarrow e^{2\pi i/3} w \Rightarrow \text{three } C_3 - \text{related orientations.}$$

The symmetry itself is primitive; the three orientations follow from the allowed cubic anisotropy. This replaces the old microscopic three-well ontology without discarding the mathematical tools developed around it. Functional renormalisation-group methods, basin analysis, cubic-anisotropy mathematics and an effective three-state representation remain useful downstream once the physical family states have been obtained from the microscopic action.

The improvement is therefore one of explanatory level. RSTM no longer requires the number three to be inserted as three independent microscopic fields, but the resulting multiplicity alone does not predict unequal masses, mixing or CP violation. Those observables require further sector-dependent dynamics. The current status is summarised in Supplementary Section O.6.

That statement remains true for C_3 multiplicity in isolation, but the required additional dynamics has now been narrowed considerably. The present action-class construction supplies an independently responding conjugate-domain reference, the fold-odd relational variable Y , the τ_2 face-family transition channel and the s -radial response through which family-relative distortion changes the matter-conditioned profiles. The old Y.12 three-well ontology is therefore not replaced merely by a count-three symmetry argument; it is replaced by a relational dynamical architecture in which hierarchy and mixing arise only after the three C_3 -related states have formed.

Prior-art and novelty boundary. Discrete C_3/Z_3 family symmetries, circulant flavour matrices and the need for additional misalignment or breaking beyond a common exact cyclic limit are not RSTM-specific [3–7]. Earlier models have used cyclic permutation symmetry and circulant matrices to organise three-family masses and mixing, while recent work has independently connected threefold discrete structures to family triplication and flavour textures [3–7]. The RSTM distinction claimed here is therefore not the cyclic flavour algebra itself. It is the origin and ordering of the family discriminator: the three states arise from the pre-existing two-real-dimensional constitutive coordinate w ; the second conjugate-domain response acts as a relational reference rather than as a second family field; the fold-odd quantity $Y = \text{Im}(w_p w_m)$ couples into the existing positive-square face sector; and the resulting coupled fixed-charge branch is selected, continued and stability-tested before a family operator is constructed. In the literature examined for the present comparison, we did not identify a construction combining this complete sequence. Its physical significance nevertheless remains conditional on blind descent of the unresolved response kernels.

4.8 Blind CKM/PMNS prediction programme

Appendix R remains useful as downstream diagonalisation and overlap machinery, but its earlier fitted three-state matrices are not retained as microscopic input. The present work now supplies the missing action-class bridge between the three constitutive orientations and the family-conditioned matter kernels.

The remaining physical inputs fall into two classes. The face splitting, u -response, profile operators and ordinary Hessian kernels must be calculated from the already-specified microscopic action. Two deeper constitutive cross-responses remain unresolved at the present level: the normalised common/radial susceptibility

$$\epsilon_{sw} = \frac{H_{sr}}{\sqrt{H_{ss}H_{rr}}} = -\frac{\chi_{sr}}{\sqrt{\chi_{ss}\chi_{rr}}}$$

and the fold-face response

$$\rho_{fold} = -\frac{\chi_{2\delta}}{\sqrt{\chi_{22}\chi_{\delta\delta}}}$$

or equivalently the stripped family susceptibility

$$\Pi_Y = c_Y^2 R H_\delta^{-1} R$$

The provenance audit is restrictive. The first response is directly expressible as a normalised bosonic susceptibility. The second is allowed by the physical diagonal fold but is not forced by fold symmetry alone, and no presently specified reflection-even RSTM mediator generates it when the face and domain odd sectors remain independently symmetric. A non-zero physical value must therefore descend from deeper fold/interface dynamics or be determined independently of flavour data.

The preferred predictive route is consequently not to fit ϵ_{sw} or Π_Y to CKM or PMNS data. Once those responses and the ordinary action-descended coefficients are fixed independently, the fixed- Q_F branch, family-conditioned profiles and inter-family saddle/overlap kernels determine M_u, M_d, M_e, M_ν . Diagonalising

$$H_f = M_f M_f^\dagger$$

then gives

$$V_{CKM} = U_u^\dagger U_d, \quad U_{PMNS} = U_e^\dagger U_\nu$$

The earlier Appendix R fits therefore remain effective demonstrations that a three-state Hermitian framework can represent the observed mixing structure. The stronger test defined here is whether independently descended RSTM responses reproduce that structure without using observed masses, CKM/PMNS elements or a fitted microscopic complex phase as branch-selection inputs. Standard flavour diagonalisation and CP diagnostics are not claimed as novel. Once the physical sector-specific matrices M_f have been obtained, the construction of $H_f = M_f M_f^\dagger$, the mismatch of left-handed diagonalising bases and rephasing-invariant tests of CP violation belong to conventional flavour theory; in particular, a physical CP claim must survive a Jarlskog-type invariant [48]. Nor is the general possibility that calculable CP violation may emerge from a symmetry-constrained theory without an arbitrary microscopic complex phase unique to RSTM [49]. The RSTM-specific test is instead whether its independently descended real fold/face responses and family-conditioned matter profiles generate a non-zero physical invariant without fitting a microscopic complex phase. The present result therefore establishes a CP-capable action-class route, not a physical CKM/PMNS or CP prediction.

4.9 Osterwalder-Schrader reconstruction and emergent Lorentz symmetry

The ultraviolet consistency result must also be read at the correct level. The stable free scalar Euclidean kernel is positive under reflection of the preferred Euclidean time and admits a positive anisotropic Hamiltonian reconstruction. Together with second-order microscopic time evolution, this is an important consistency check on the selected higher-spatial-derivative action.

It is not, however, the standard Lorentz-invariant Osterwalder-Schrader theorem. At the microscopic $z=3$ scaling point, Euclidean time and space are anisotropic. A complete relativistic reconstruction requires a separate long-wavelength trajectory on which the second-order spatial and temporal terms dominate and the physical matter and gravitational sectors acquire the same Lorentz cone.

preferred-time UV theory $\rightarrow$ infrared Lorentz-restoring trajectory $\rightarrow$ relativistic QFT sector.

The same distinction applies to spin-statistics. The reduced spin and exchange constructions provide the required precursors, but the usual Lorentz-covariant theorem [47] becomes applicable only after the local $\text{Spin}(1,3)$ field, positivity and microcausality conditions have been obtained. The ultraviolet and reflection-positivity audit is given in Supplementary Section L; the continuity boundary is summarised in Supplementary Section O.9.

4.10 Ultraviolet preferred-frame memory

Emergent Lorentz symmetry does not require the microscopic substrate to erase all memory of its preferred clock or rest frame at every scale. The stronger requirement is that the leading physical infrared theory lose observable dependence on an absolute spatial ruler and recover the required Lorentz/Einstein structure. Higher-derivative operators may then retain information about the microscopic substrate.

The reference-erasure principle should not itself be interpreted as an RSTM-exclusive route to linearised gravity. Closely related compatibility/gauge reasoning appears in continuum and discrete gravitational constructions, most directly in the recent intrinsic D_4 Regge model [32]. The distinction sought here is the microscopic provenance of the collective field and of its kernel. RSTM does not begin with intrinsic gravitational edge lengths or assume a Fierz-Pauli field: its target spatial metric is assembled from collective moments of the underlying substrate, and the Einstein coefficient ratio is recovered only after compatible ruler directions are required to decouple from that collective response. Whether the same microscopic action also supplies the healthy temporal completion, physical tensor pole, common propagation cone and nonlinear constraint algebra remains an open test rather than being inferred from the spatial Ward result.

$$\Delta K_{ref}(k) = 0 + 0 \cdot k^2 + \alpha_4 k^4 + \alpha_6 k^6 + \dots$$

The first two terms in this schematic hierarchy express the absence of homogeneous reference pinning and placement of the two-derivative spatial sector on the Ward-protected surface. The higher-order terms are not required to vanish. RSTM may therefore predict small scale-dependent preferred-frame or dispersive effects without contradicting its infrared target. Conversely, taking k to zero does not automatically repair an incorrect leading operator: the collective dynamics must genuinely land on the Ward/Lorentz surface. The corresponding gravitational hierarchy is discussed in Supplementary Section K and the ultraviolet operator constraints in Supplementary Section L.

4.11 Implications for calibration-first quantum control

The revised microscopic parameter discipline does not invalidate calibration-first quantum-control strategies. A device-control protocol may legitimately infer effective frequencies, couplings, phases

or decoherence parameters from experiment and use them to optimise control without knowing their ultimate microscopic origin. Such a method is operational rather than ontological.

What the present work excludes is the reverse inference in which successful downstream calibration is presented as a first-principles derivation of J^* , microscopic stiffnesses or other fundamental RSTM coefficients. The control architecture can therefore remain theory-independent while the microscopic programme proceeds separately. If RSTM eventually derives those coefficients, the result becomes an additional predictive constraint on the calibrated control model rather than an assumption used to build it.

4.12 Falsification programme

The microscopic reconstruction makes the programme more falsifiable because several former fit parameters have become correlated outputs or blind comparison targets. The most direct failures would be:

- action-derived physical scalar coefficients whose exact Schur-reduced kernel fails to remain positive while developing the required finite- k preferred minimum; where the controlled local $2 - 4 - 6$ reduction applies, failure to recover its corresponding corridor and scale-selection behaviour;
- physical face coefficients that fail the global positive-parent conditions or do not support a fixed-charge one-orbital branch below the dispersed same-charge continuum;
- loss of the radial, non-radial or dormant-face stability corridor when benchmark coefficients are replaced by coefficients descended from the microscopic action;
- failure of the normalised phase-frame sector to descend microscopically or to coexist stably and centre-lock with the physical fixed-charge occupation branch;
- failure to obtain a universal elementary action normalisation whose blind downstream comparison is compatible with $\hbar$;
- failure of the complete microscopic dynamics to select the already-available odd rank-two matter sector, or failure of that sector to admit a local Lorentz-covariant and microcausal field completion;
- failure of the structural internal-frame connection and charge classes to descend to a physical local gauge redundancy with the required kinetic normalisation and couplings;
- failure of the physical common tensor response to produce a healthy massless infrared pole, positive residue, the required common cone and the spatial Einstein/Ward structure;
- failure of the independently obtained physical values of $\downarrow_*$, J_* and G_{RSTM} to satisfy the normalisation-independent cross-sector closure.
- failure of the microscopic theory to generate a non-zero family connector — $\epsilon_{sw} = 0$ would prevent family-radial response from reaching the common constitutive/matter sector, while $\Pi_Y = 0$ or $\rho_{fold} = 0$ would leave the face and family odd sectors dynamically disconnected; likewise, failure of the physically descended coefficients to produce a stable nonlinear branch, or blind family operators that remain simultaneously diagonal or give mass/mixing structures incompatible with observation, would falsify the corresponding flavour construction;

Ultraviolet Lorentz violation is not, by itself, a falsification of the model because preferred-frame memory is allowed in higher-derivative terms. The sharper test is failure to recover the required Lorentz and Einstein structure in the leading physical infrared sector, or the appearance of ultraviolet corrections inconsistent with the coefficients derived by the theory. The detailed continuity of the earlier experimental proposals is discussed in Supplementary Section O, Section O.11.

4.13 Remaining open gates

The present work therefore closes some questions while making the remaining ones more specific. The principal open gates are:

- derivation of the physical face coefficients from the microscopic action and blind testing of those coefficients against the already-established fixed- Q_F , one-orbital localisation and stability corridor;
- derivation of the absolute elementary action scale J_* and its blind comparison with $\hbar$;
- microscopic descent of the already-constructed normalised $B=1$ phase-frame bundle from the physical face/bundle and interaction action within the accepted RSTM microscopic architecture, followed by a fully action-derived, centre-locked amplitude-frame solution on the physical fixed- Q_F carrier;
- physical derivation that realistic RSTM preparation dynamics universally samples the already-identified invariant Haar/Liouville phase measure, together with proof that the resulting detector-basin measures reproduce the quadratic channel weights for arbitrary admissible measurement contexts;
- derivation of the microscopic elementary-occupancy and branch-selection rule that compulsorily selects the already-established rank-two matter sector carrying the reduced centre-odd spin structure and, independently, the established exchange-odd representation, followed by its promotion to a local $\text{Spin}(1,3)$ -covariant, particle/antiparticle-complete and graded-microcausal fermion field;
- descent of the already-established internal-frame connection and global charge-class structure to local physical gauge redundancy, together with action-derived Yang-Mills kinetic normalisation, Ward identities and physical gauge couplings;
- derivation from the microscopic action of physical family operators on the already-established three $\mathbf{C}_3$ -related internal states, yielding the physical mass hierarchy, CKM/PMNS mixing and CP structure without fitted family inputs;
- derivation of the physical projection coefficients mapping the already-available matter stress/texture structure of the solved fixed- Q_F carrier onto the collective gravitational source channels, thereby completing the effective energy-momentum source map;
- simultaneous dynamical realisation in the actual closed RSTM tensor kernel of the already-identified common-channel requirements: a single healthy massless transverse-traceless pole, positive residue, common infrared propagation cone, vanishing leading infrared width, a gapped relative tensor channel and metric-normalised G_{RSTM} ;

- microscopic descent of the already-specified soft Fierz-Pauli recombination target from the relaxed scalar/vector/tensor response and completion of the physical collective metric/source map;
- derivation of a temporal infrared operator-selection or Lorentz-naturalness mechanism protecting the completed infrared surface against dangerous preferred-frame operators and radiative destabilisation;
- derivation of the RSTM collective Hamiltonian and momentum constraints and closure of their hypersurface-deformation algebra, thereby completing the structurally known route from HDA closure to DeWitt/ADM and Einstein dynamics;
- physical validation and selection of the already-derived M1 relaxation length of Eq. (N.4) as the universal microscopic crossover/ruler scale ℓ_{-}^* , including the required blind carrier/similarity-collapse tests, followed by evaluation of the normalisation-independent cross-sector Planck-closure condition;
- transfer from microscopic/core regularity to a regular, non-degenerate collective metric through the regulated core, followed by completion of the strong-field compact-object geometry, including curvature control, horizon structure and causal/geodesic completion;
- derivation of ultraviolet mode conversion on the completed strong-field geometry and its connection to the correct infrared Hawking behaviour;
- an independent microscopic horizon-state count after the physical constraints, reference quotient and boundary conditions have been imposed;
- complete coupled RSTM ultraviolet closure beyond the selected finite scalar/constitutive truncation;
- derivation of the vacuum/conjugate-sector effective stress, including the physical homogeneous residual, equation of state and radiative stability, after the infrared gravitational sector has been completed.

The existence of a benchmark persistent three-dimensional matter carrier should not be included in this list as an undifferentiated open problem. At the claim boundary, an explicit positive-square face-action class already supports a finite one-orbital fixed-Noether-charge localised branch with radial, non-radial and full-face stability controls. The remaining particle-level problem is therefore not the existence of such a carrier, but the physical descent of its coefficients and normalised phase-frame structure, together with the microscopic occupancy and sector-selection rules that determine which member of the available stable branch structure is physically elementary.

The resulting position is narrower than the broadest claims of the earlier RSTM formulation but materially stronger at the microscopic level. Stable finite-scale structure, a conservative fixed-charge matter carrier, a reduced spin/exchange architecture, positive-frequency quantum-like evolution and a non-trivial collective spatial geometry now arise within one action-first hierarchy. The next stage is therefore not to add further phenomenological sectors, but to determine whether the same microscopic action fixes the remaining dimensional normalisations, compulsory matter/gauge selections and full four-dimensional gravitational dynamics. Supplementary Section O gives the detailed continuity audit and Supplementary Section P fixes the claim-status language used for that programme.

5. Conclusion

This paper has replaced the earlier picture of RSTM as a single phenomenological master equation with an action-first hierarchy in which effective, reduced and structural objects are organised at different levels within a conservative microscopic architecture. The minimal constitutive ontology remains four-dimensional and contains the scalar membrane coordinate u together with the internal variables s and w . From that starting point, constitutive relaxation, fixed-charge selection, positive-frequency reduction, open-system projection and collective reference construction supply different downstream structures without requiring those structures to be inserted independently as microscopic laws.

The principal outcomes may be summarised as Structure, Scale, Particle, Family, Quantum and Geometry. The effective scalar hierarchy is descended from a positive microscopic constitutive parent; a stable finite structural scale is dynamically selected; a distinct fixed-charge mechanism supplies an existence-class localised matter carrier; a relational conjugate-domain response supplies an action-class route from three C_3 orientations to stable family-conditioned matter kernels; positive-frequency reduction produces the complex quantum-like description and its carrier action; and incompatible collective strain supplies the spatial Einstein/Ward geometry. The normalisation-independent cross-sector invariant then provides a common blind test of whether the independently determined physical values of the microscopic length, action and gravitational sectors converge on their observed counterparts.

The first principal result is structural. On a curvature-populated branch, elimination of a healthy common constitutive coordinate produces intermediate-gradient softening while simultaneously increasing the sixth-order regulator,

$$\Delta K_4 < 0, \quad \Delta K_6 > 0, \quad \Delta K_6 = -\downarrow_s^2 \Delta K_4.$$

The corresponding reduced scalar kernel possesses a stable finite-wave-number corridor and a narrow parameter-independent wavelength interval relative to its intrinsic sixth-order length. The familiar RSTM second-, fourth- and sixth-order scalar equation therefore survives, but as a collective law whose coefficients are correlated outputs of microscopic equilibrium and constitutive reduction rather than independently fitted fundamental constants.

The second principal result is the separation of scale selection from particle binding. The finite-wave-number scalar minimum selects a preferred structural scale but does not itself provide a persistent sub-continuum scalar particle. Persistent localisation instead occurs in the conservative two-face occupation sector. An exact common phase symmetry supplies a Noether charge Q_F , and reciprocal coupling to the common constitutive coordinate supports a finite benchmark interval containing one preferred spatial orbital below the dispersed same-charge continuum. The branch survives the tested radial, non-radial and complementary-face stability audits. At the claim boundary, the existence of a benchmark three-dimensional matter carrier is therefore established for the declared action class; the remaining task is to derive the physical coefficients and determine whether the microscopic theory selects that branch.

The third principal result is the family/flavour construction. The primitive C_3 sector gives exactly three internal orientations but does not by itself lift their degeneracy. Domain-resolved relational response and the existing τ_2 face/ u channel provide the additional action-class structure required to distinguish their matter-conditioned responses. The resulting fixed-charge odd branch has been continued nonlinearly and remains stable through the tested radial, non-radial and co-rotating sectors. This establishes a viable microscopic route from three constitutive orientations to family-dependent matter kernels without inserting a family matrix or microscopic complex Yukawa phase. Physical fermion masses and CKM/PMNS matrices are not claimed as results of the present paper; they require physical descent of the remaining response kernels followed by sector-specific diagonalisation.

The matter construction also clarifies which structures must remain distinct. Binding resides in the

fixed-charge occupation amplitude, while a separately normalised and covariantly attached phase frame carries the double-cover spin topology. The resulting rank-two positive-frequency fibre has the characteristic sign reversal under a 2π rotation. An independent two-particle construction gives reduced exchange oddness, from which the standard mode canonical anticommutation relations follow conditionally once the odd sector is physically selected. At the present structural level, chirality and internal-frame gauge architecture are supplied through separate face/orientation and covariance mechanisms. Their local relativistic and Yang-Mills completions remain open. The reduced quantum-like description is obtained downstream of the real conservative face dynamics. Positive-frequency selection produces Schrödinger-like complex evolution, while the exact face-phase Noether charge coincides structurally with the spectral and symplectic carrier actions on the circular branch. The elementary dimensional action scale J_-^* is not fixed by classical orbit geometry and has therefore not been set equal to $\hbar$. Quadratic detector response produces Born-form channel weights and the representative deterministic threshold model demonstrates a corresponding one-outcome recording mechanism. The positive-frequency orbit also supplies a natural invariant Haar/Liouville phase measure as a candidate unresolved-state measure:

$$d\mu = \frac{d\theta}{2\pi}$$

The remaining statistical problem is therefore not the absence of a candidate measure, but the physical derivation that realistic preparation dynamics samples this invariant measure universally and that the resulting detector-basin measures reproduce the quadratic channel weights for arbitrary admissible measurement contexts. Until that preparation theorem is obtained, the present detector result remains representative rather than a universal Born-rule derivation.

The gravitational result is similarly structural but non-trivial. The primitive scalar field is not identified directly with the metric. A scalar plus traceless-quadrupole collective sector supplies the six components of a spatial metric, compatible strain is interpreted as reference or ruler deformation, and incompatible collective strain reproduces the linearised spatial Einstein tensor. Spatial relabelling symmetry then fixes the leading two-derivative curvature coefficients to the Einstein ratio. The remaining gravitational problem is therefore not the form of the required spatial or relativistic structure, whose principal targets are already fixed, but their simultaneous physical descent from the microscopic action. This requires the matter-source projection, a single healthy common transverse-traceless spectral branch with the required pole, residue and propagation cone, metric normalisation, soft temporal Fierz-Pauli recombination, infrared operator protection and nonlinear collective constraint closure.

The same action retains second-order microscopic time evolution while positive sixth-order spatial terms provide $z=3$ ultraviolet control for the selected finite truncation and an energetic obstruction to zero-width collapse. The retained core calculation exhibits leading area-law behaviour at its stated reduced level. These results do not yet constitute complete coupled renormalisability, an all-scale Lorentz-invariant Osterwalder-Schrader theorem, strong-field metric regularity or an independent microscopic horizon-state count.

A final consistency condition links the otherwise independently defined microscopic length, action and gravitational sectors. Their normalisation-independent combination is

$$C_* = \frac{c_{\downarrow*}^3}{J_* G_{RSTM}} = \frac{8\pi\zeta_\varepsilon}{\hat{J}_*}.$$

Because this quantity is invariant under an overall rescaling of the classical microscopic action, it provides a genuine blind cross-sector test rather than a restatement of Planck units. The physical values of ℓ_-^* , J_-^* and G_{RSTM} must first be derived independently. Only then can the theory be compared with the conventional Planck length, $\hbar$ and Newton's constant.

For the length sector, this does not mean that no microscopic candidate is presently available: the constitutive dynamics already selects the M1 relaxation length defined in Eq. (N.4) as the

preferred candidate. What remains is its blind physical validation as the universal ruler ℓ_{-}^* , rather than construction of a length scale from scratch.

The six principal results of the present construction may therefore be summarised as Structure, Scale, Particle, Family, Quantum and Geometry. Their common feature is not that familiar physics has been assumed in six different forms, but that each is situated within the same conservative microscopic architecture through an explicitly stated combination of dynamics, physical selection, reduction or collective construction. This is the central sense in which the current formulation advances beyond the earlier effective RSTM model.

The next stage is correspondingly well defined. The decisive question is whether the same microscopic action can determine the absolute elementary action, the physical face, family-response and frame coefficients, compulsory fermion and gauge selection, and the full four-dimensional gravitational response without importing the observed quantities that those calculations are intended to explain. Success would turn the present existence and structural results into a predictive microscopic theory; failure of any of the blind coefficient, localisation, Lorentz/Einstein or cross-sector closure tests would falsify the corresponding RSTM construction.

Statements

- **Author contribution:** The author confirms the sole responsibility for the conception of the study, development of the STM model, analysis of the results and preparation of the manuscript.
- **Funding information:** The author received no specific funding for this work.
- **Data availability:** All data generated or analysed during this study are included in this published article and its supplementary information files.
- **Acknowledgements** I would like to express my deepest gratitude to the scholars and researchers whose foundational work is cited in the references; their contributions have been instrumental in the development of the Space-Time Membrane (STM) model presented in this paper. I am thankful for the advanced computational tools and language models that have supported the mathematical articulation of the STM model, which I have developed over the past seventeen years. Finally, I wish to pay tribute to my mother, Mavis, for my tenacity and resourcefulness; my father, James, for my imagination; my wife and children, Joanne, Elliot and Louis, for their belief in me; and to the late Isaac Asimov, whose writings first sparked my enduring curiosity in physics.
- **Conflict of interest:** The author declares no conflict of interest.
- **Ethical approval:** The conducted research is not related to either human or animals use.
- **Declaration of generative AI and AI-assisted technologies in the writing process:** During the preparation of this work the author used ChatGPT in order to improve readability of the paper and support with mathematical derivations. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Supplementary Information

Supplementary Sections A-P provide the detailed derivations, numerical stability audits, continuity analysis, notation and claim-status definitions supporting the main text. Supplementary Figures S1-S4 provide additional diagnostics for the nonlinear/flavour-branch numerical cross-checks, direct radial stability of the fixed-Noether-charge carrier, translation-mode convergence, and

complementary-face Schur coercivity, respectively. Supplementary Simulation S1 visualises the perturbed radial evolution of the oscillating real matter quadratures and the persistence of the invariant localised carrier. Supplementary Code S1–S5 provides the reproducibility package for the local finite-scale corridor, wavelength calculation and exact-rational-kernel consistency control (S1), fixed-Noether-charge particle branch and stability audits (S2), deterministic threshold-detector benchmark (S3), collective Einstein/Ward symbolic checks (S4), and family/flavour action-class closure and response-descent audits (S5). The code package includes source code, numerical provenance and settings, reference outputs, manuscript-to-code mapping and automated validation checks.

References

- [1] Swann, P. The Refined Space-Time Membrane Model: Deterministic Emergence of Quantum Fields and Gravity from Classical Elasticity, Version 7. Preprints 2026, 202503.0736.v7. doi:10.20944/preprints202503.0736.v7.
- [2] Zhang, F., Ed. The Schur Complement and Its Applications; Numerical Methods and Algorithms, Vol. 4; Springer: New York, 2005. doi:10.1007/b105056.
- [3] Adler, S. L. Model for Particle Masses, Flavor Mixing, and CP Violation Based on Spontaneously Broken Discrete Chiral Symmetry as the Origin of Families. Phys. Rev. D 1998, 59, 015012. doi:10.1103/PhysRevD.59.015012.
- [4] Harrison, P. F.; Scott, W. G. Symmetries and Generalisations of Tri-Bimaximal Neutrino Mixing. Phys. Lett. B 2002, 535, 163-169. doi:10.1016/S0370-2693(02)01753-7.
- [5] Patrascu, A. T. Coherence Obstructions and Family Triplication: A Categorical Mechanism for Three Fermion Generations. Eur. Phys. J. C 2026, 86, 703. doi:10.1140/epjc/s10052-026-15900-z.
- [6] Akhtman, Y.; Voether, E. The Fermion Spectrum over Finite Relational Substrate, Version 1. Preprints 2026, manuscript 202606.2124. [Preprint; not peer reviewed.]
- [7] Bazih, A. Three Fermion Generations from Z_3 Triality in Modular Tensor Categories. Zenodo 2026, Version v1. doi:10.5281/zenodo.19548222. [Preprint; not peer reviewed.]
- [8] Brazovskii, S. A. Phase transition of an isotropic system to a nonuniform state. Sov. Phys. JETP 1975, 41, 85-89.
- [9] Swift, J.; Hohenberg, P. C. Hydrodynamic fluctuations at the convective instability. Phys. Rev. A 1977, 15, 319-328. doi:10.1103/PhysRevA.15.319.
- [10] Noether, E. Invariant variation problems. Transport Theory Stat. Phys. 1971, 1, 186-207. doi:10.1080/00411457108231446. English translation of the 1918 paper.
- [11] Coleman, S. Q-balls. Nucl. Phys. B 1985, 262, 263-283. doi:10.1016/0550-3213(85)90286-X.
- [12] Leese, R. A. Q-Lumps and their interactions. Nucl. Phys. B 1991, 366, 283-311. doi:10.1016/0550-3213(91)90004-H.
- [13] Volkov, M. S.; Wöhnert, E. Spinning Q-balls. Phys. Rev. D 2002, 66, 085003. doi:10.1103/PhysRevD.66.085003.
- [14] Adam, C.; Naya, C.; Sanchez-Guillen, J.; Wereszczynski, A. Lifshitz field theories with SDiff symmetries. J. High Energy Phys. 2013, 03, 012. doi:10.1007/JHEP03(2013)012.

- [15] Finkelstein, D.; Rubinstein, J. Connection between spin, statistics, and kinks. *J. Math. Phys.* 1968, 9, 1762-1779. doi:10.1063/1.1664510.
- [16] Laidlaw, M. G. G.; Morette DeWitt, C. Feynman functional integrals for systems of indistinguishable particles. *Phys. Rev. D* 1971, 3, 1375-1378. doi:10.1103/PhysRevD.3.1375.
- [17] Born, M. Zur Quantenmechanik der Stossvorgaenge. *Z. Phys.* 1926, 37, 863-867. doi:10.1007/BF01397477.
- [18] Boehmer, C. G.; Lee, Y. Compatibility conditions of continua using Riemann-Cartan geometry. *Math. Mech. Solids* 2021, 26, 513-529. doi:10.1177/1081286520961453.
- [19] Fierz, M.; Pauli, W. On relativistic wave equations for particles of arbitrary spin in an electromagnetic field. *Proc. R. Soc. Lond. A* 1939, 173, 211-232. doi:10.1098/rspa.1939.0140.
- [20] Hojman, S. A.; Kuchar, K.; Teitelboim, C. Geometrodynamics regained. *Ann. Phys.* 1976, 96, 88-135. doi:10.1016/0003-4916(76)90112-3.
- [21] Horava, P. Quantum gravity at a Lifshitz point. *Phys. Rev. D* 2009, 79, 084008. doi:10.1103/PhysRevD.79.084008.
- [22] Anselmi, D. Weighted power counting and Lorentz violating gauge theories. I: General properties. *Ann. Phys.* 2009, 324, 874-896. doi:10.1016/j.aop.2008.12.005.
- [23] Woodard, R. P. Ostrogradsky's theorem on Hamiltonian instability. *Scholarpedia* 2015, 10(8), 32243. doi:10.4249/scholarpedia.32243.
- [24] Osterwalder, K.; Schrader, R. Axioms for Euclidean Green's functions. *Commun. Math. Phys.* 1973, 31, 83-112. doi:10.1007/BF01645738.
- [25] Osterwalder, K.; Schrader, R. Axioms for Euclidean Green's functions II. *Commun. Math. Phys.* 1975, 42, 281-305. doi:10.1007/BF01608978.
- [26] Bekenstein, J. D. Black holes and entropy. *Phys. Rev. D* 1973, 7, 2333-2346. doi:10.1103/PhysRevD.7.2333.
- [27] Wald, R. M. Black hole entropy is the Noether charge. *Phys. Rev. D* 1993, 48, R3427-R3431. doi:10.1103/PhysRevD.48.R3427.
- [28] Hawking, S. W. Particle creation by black holes. *Commun. Math. Phys.* 1975, 43, 199-220. doi:10.1007/BF02345020.
- [29] Molzberger, L. A Deterministic Brane-Lattice Substrate for Emergent Spacetime, Gauge Structure, and Matter. *Research Square* 2026. doi:10.21203/rs.3.rs-10028872/v1.
- [30] Jiang, K.-X. Isomorphic Emergence of Lorentz and Gauge Symmetries--A Constructive Interpretation Based on Continuum Mechanics. *arXiv* 2026, arXiv:2608.06244.
- [31] Singh, M. Quantum Elastic Spacetime Theory (QuEST): A Strain-Saturated Framework for Singularity Avoidance, Gravitational Echoes, and Emergent Geometry. *Zenodo* 2025, Version v2. doi:10.5281/zenodo.15768085. [Preprint; not peer reviewed.]
- [32] Kulkarni, R. Emergent Linearized Gravity from the Intrinsic Lattice: the Graviton as the Incompatible Edge-Length Mode. 2026. doi:10.5281/zenodo.20144486.
- [33] Boyle, L.; Finn, K.; Turok, N. CPT-Symmetric Universe. *Phys. Rev. Lett.* 2018, 121, 251301. doi:10.1103/PhysRevLett.121.251301.

- [34] Boyle, L.; Teuscher, M.; Turok, N. The Big Bang as a Mirror: a Solution of the Strong CP Problem. arXiv 2022, arXiv:2208.10396.
- [35] Zwanzig, R. Memory effects in irreversible thermodynamics. *Phys. Rev.* 1961, 124, 983-992. doi:10.1103/PhysRev.124.983.
- [36] Mori, H. Transport, collective motion, and Brownian motion. *Prog. Theor. Phys.* 1965, 33, 423-455. doi:10.1143/PTP.33.423.
- [37] Swann, P. The Refined Space-Time Membrane Model: Deterministic Emergence of Quantum Fields and Gravity from Classical Elasticity, Version 3. Preprints 2025, 202503.0736.v3. doi:10.20944/preprints202503.0736.v3.
- [38] Singh, P. Origin of Complex and Spinor Structures in Quantum Mechanics from Real Second-Order Dynamics. SSRN 2026. doi:10.2139/ssrn.6735029.
- [39] Tong, D. Line operators in the Standard Model. *J. High Energy Phys.* 2017, 07, 104. doi:10.1007/JHEP07(2017)104.
- [40] Yang, C. N.; Mills, R. L. Conservation of isotopic spin and isotopic gauge invariance. *Phys. Rev.* 1954, 96, 191-195. doi:10.1103/PhysRev.96.191.
- [41] Khrennikov, A. Born's rule from measurements of classical signals by threshold detectors which are properly calibrated. *J. Mod. Opt.* 2012, 59, 667-678. doi:10.1080/09500340.2012.656718.
- [42] Goldstone, J.; Salam, A.; Weinberg, S. Broken symmetries. *Phys. Rev.* 1962, 127, 965-970. doi:10.1103/PhysRev.127.965.
- [43] Derrick, G. H. Comments on nonlinear wave equations as models for elementary particles. *J. Math. Phys.* 1964, 5, 1252-1254. doi:10.1063/1.1704233.
- [44] Kielhöfer, H. Bifurcation Theory: An Introduction with Applications to Partial Differential Equations, 2nd ed.; Applied Mathematical Sciences, Vol. 156; Springer: New York, 2012. doi:10.1007/978-1-4614-0502-3.
- [45] Bouchiat, C.; Iliopoulos, J.; Meyer, P. An anomaly-free version of Weinberg's model. *Phys. Lett. B* 1972, 38, 519-523. doi:10.1016/0370-2693(72)90532-1.
- [46] Witten, E. An SU(2) anomaly. *Phys. Lett. B* 1982, 117, 324-328. doi:10.1016/0370-2693(82)90728-6.
- [47] Pauli, W. The connection between spin and statistics. *Phys. Rev.* 1940, 58, 716-722. doi:10.1103/PhysRev.58.716.
- [48] Jarlskog, C. Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation. *Phys. Rev. Lett.* 1985, 55, 1039-1042. doi:10.1103/PhysRevLett.55.1039.
- [49] Branco, G. C.; Gérard, J.-M.; Grimus, W. Geometrical T-Violation. *Phys. Lett. B* 1984, 136, 383-386. doi:10.1016/0370-2693(84)92024-0.
- [50] Woodhouse, N. M. J. Geometric Quantization, 2nd ed.; Oxford Mathematical Monographs; Clarendon Press/Oxford University Press: Oxford, 1992. doi:10.1093/oso/9780198536734.001.0001.
- [51] Bell, J. S. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika* 1964, 1, 195-200. doi:10.1103/PhysicsPhysiqueFizika.1.195.

- [52] Iyer, V.; Wald, R. M. Some properties of the Noether charge and a proposal for dynamical black hole entropy. *Phys. Rev. D* 1994, 50, 846-864. doi:10.1103/PhysRevD.50.846.
- [53] Unruh, W. G. Sonic analogue of black holes and the effects of high frequencies on black hole evaporation. *Phys. Rev. D* 1995, 51, 2827-2838. doi:10.1103/PhysRevD.51.2827.
- [54] Corley, S.; Jacobson, T. Hawking spectrum and high frequency dispersion. *Phys. Rev. D* 1996, 54, 1568-1586. doi:10.1103/PhysRevD.54.1568.